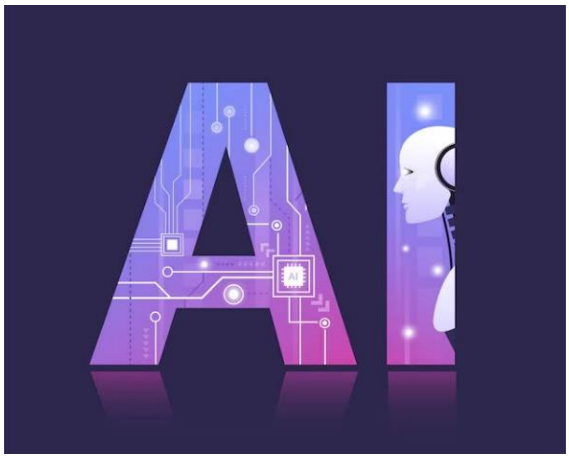
Degree Project in Innovation Management and Product Development

Second cycle, 30 credits

Factors Influencing the Utilisation and Adoption of AI Chatbots in Organisational Environments

A qualitative & quantitative analysis of the use and adoption of an AI chatbot, Pixie, at Piab Group AB

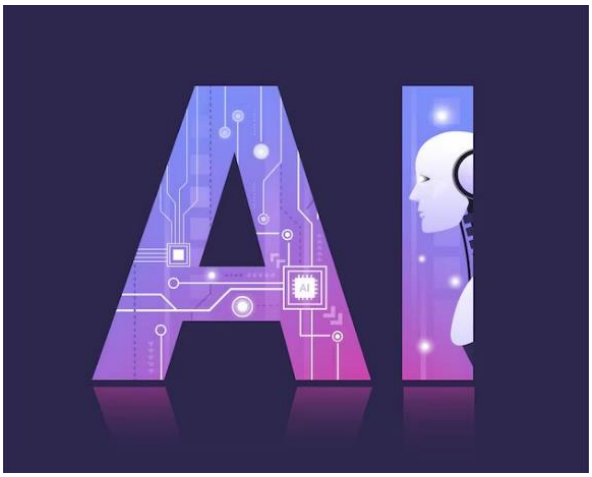
PRANEET KALA & SANJAY RAJASEKARAN



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Master of Science Thesis TRITA-ITM-EX 2024:524
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Faktorer som påverkar användning och införande av AI-chatbotar i organisationer

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Sammanfattning

Denna forskning undersöker faktorer som påverkar användningen och införandet av AI-chatbots inom organisationer. Genom empiriska studier inom ett teknikdrivet företag syftar studien till att förstå hur man effektivt kan implementera och utnyttja AI-chatbots för att öka affärsvärdet. En metodansats med både kvalitativa intervjuer och kvantitativa undersökningar används för att samla in insikter från anställda från olika avdelningar. Studien identifierar nyckelfaktorer för användning och införande av AI-chatbots och betonar vikten av tillräcklig utbildning, borttagning av barriärer och ökad medvetenhet om verktygets kapacitet. Kulturellt motstånd, begränsad relevans för arbetsroller och behovet av intern integration med externa applikationer framstår också som betydande utmaningar. Dessutom visar resultaten på vikten av att förbättra AI-chatbot-funktioner som uppgiftsautomatisering, bildbehandling och kritiska prestandafaktorer som tillförlitlighet och noggrannhet. Resultatet från detta examensarbete bidrar till förståelsen av AI-teknologins adoption på arbetsplatsen och ger praktiska rekommendationer för att optimera användningen av AI-chatbots.

Nyckelord

AI, ChatGPT, AI på arbetsplatsen, AI-produktutveckling, Piab Group, Pixie AI, Kvalitativ & Kvantitativ studie



KTH Industrial Engineering and Management

Master of Science Thesis TRITA-ITM-EX 2024:524

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Abstract

This research investigates the factors influencing the utilisation and adoption of chatbots within organisations. By examining a technology-first company, the study aims to understand how to effectively implement and leverage chatbots to maximise business value. A mixed-methods approach, combining qualitative interviews and quantitative surveys, gathers insights from employees across different departments. The study identifies key determinants of chatbot usage and adoption, highlighting that adequate training, removing adoption barriers, and improving tool capability awareness are crucial factors. Cultural resistance, limited relevance to job roles, and the need for internal integration with external applications also emerge as significant challenges. Additionally, the study underscores the importance of enhancing chatbot features, such as task automation, image processing, and critical performance factors like reliability and accuracy. This research contributes to understanding AI technology adoption in the workplace and provides practical recommendations for optimising AI chatbots.

Keywords

AI, ChatGPT, AI at Work place, AI Product Development, Piab Group, Pixie AI, Qualitative & Quantitative study

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Lastly, we would like to thank our friends and family for all their support during this thesis project and our entire education.

Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
CRM	Customer Relationship Management
ERP	Enterprise Resource Planning
ERM	Enterprise Resource Management
FAQ	Frequently asked questions
FTE	Full-Time Employment
GDIS	Global Digital and Information Systems
GM	General Management
GUI	Graphical User Interface
HR	Human Resources
IT	Information Technology
LA	Lifting Automation
LLM	Large Language Models
ML	Machine-Learning
MS	Microsoft
MT	Machine Translation
NLP	Natural Language Processing
NPS	Net Promoter Score
RAG	Retrieval-Augmented Generation
R&D	Research & Development
RQ	Research Question
UI	User Interface
VA	Vacuum Automation
VC	Vacuum Conveying

Table of Contents

1 Introduction **1**

1.1 Background and Problem . . . 1

1.1.1 Piab Group . . . 1

1.2 Purpose . . . 4

1.3 Goal . . . 4

1.4 Benefits, Ethics and Sustainability . . . 5

1.4.1 Benefits . . . 5

1.4.2 Ethics . . . 6

1.4.3 Sustainability . . . 7

1.5 Methodology . . . 7

1.6 Stakeholders . . . 8

1.7 Delimitation's . . . 8

1.8 Outline . . . 9

2 Literature Review **10**

2.1 Artificial Intelligence . . . 10

2.2 Chatbots . . . 12

2.3 Retrieval-Augmented Generation Models . . . 13

2.4 Business Case . . . 14

2.5 Improving usage of AI chatbots . . . 16

3 Method **19**

3.1 Research Design . . . 19

3.2 Phase 1 - Data Sorting and Interviewee Selection Process . . . 20

3.2.1 User Data Analysis/Data Mining . . . 24

3.3 Phase 2 - Data collection through interviews . . . 25

3.3.1 Interview Guide . . . 25

3.3.2 Interview Candidates . . . 27

3.3.3 Interview Coding . . . 28
3.4 Phase 3 - Data collection through survey . . . 29
3.4.1 Survey . . . 29
3.5 Phase 4 - Recommendations and Refinement . . . 34
3.6 Methodological assessment . . . 34
3.7 Research Bias . . . 35

4 Empirical Findings 39
4.1 Power BI Dashboards . . . 39
4.2 Interviews . . . 41
4.2.1 General findings . . . 41
4.2.2 User Interaction and Experience with Automated Tools . . . 43
4.2.3 Training, Challenges, and Impact on Job role . . . 45
4.2.4 Future Enhancements and Digital Transformation . . . 48
4.3 Survey . . . 49
4.3.1 Overall User experience . . . 51
4.3.2 Usage Frequency and Purpose . . . 52
4.3.3 Productivity . . . 52
4.3.4 Feature requests . . . 53
4.3.5 Performance . . . 54
4.3.6 Challenges . . . 56
4.3.7 Training and its Effectiveness . . . 57

5 Analysis 59
5.1 Interview Analysis . . . 59
5.1.1 Info from Usage Patterns . . . 60
5.1.2 Info from User Experience . . . 60
5.1.3 Info from Impact and Value . . . 61
5.1.4 Info from Feedback . . . 61
5.2 Function specific insights into results . . . 62
5.2.1 Feature utilisation . . . 63
5.2.2 Productivity . . . 64
5.2.3 Feature requests . . . 65
5.2.4 Challenges . . . 66

6 Discussion 68

6.1 Overview 68
6.2 Interpretation of findings 69
6.2.1 Answer to RQ 1 69
6.2.2 Answer to RQ 2 72

7 Conclusion **76**
7.1 Implication of findings 76
7.2 Limitations of the study 77
7.3 Future research directions 77

References **78**

Appendices **i**

A Interview Guide **ii**

B Project Plan **iv**

C Additional Power BI Charts **v**

D Interview Coding **vii**

Introduction

The introduction chapter of this project provides an overview of the background and research problem. The background is followed by the purpose of the research and the importance and significance of the study with the research questions. A short description of the goal of the project is described. benefits, ethics and sustainability topics that are relevant are introduced and the methodology used in the project is briefly discussed. The stakeholders involved are mentioned and the section concludes with delimitation's and report outline.

1.1 Background and Problem

The prevalence of Chatbots in corporate settings is steadily rising, primarily driven by their capacity to reduce operational expenses associated with customer support and concurrently handle a diverse array of users efficiently (Ali et al., 2024; Hsu et al., 2023). What changed the game was the Retrieval-Augmented Generation (RAG) model that significantly reduces the hallucination in the provided results, can base its results on a wide array of organisation specific data and documents and makes the data accessible & safe in the cloud. The ongoing growth of the Industry 4.0 trend, advancements in technology, and the rise of Artificial Intelligence (AI), now provides organisations the ability to make extraordinary progress at unprecedented rates.

The question with any new implementation of such tools in an organisational and enterprise context boils down to why & how. Why AI chatbots should be introduced in an organisation and how it could be introduced and implemented.

1.1.1 Piab Group

Piab was founded in 1951 and has been an evolving industrial automation organisation. They advocate for an automated world where resources are utilised efficiently, and the well-being of humans remains safeguarded with no injuries. They improve customers' productivity, sustainability, and safety in all industries globally, with their automated process components. Special attention is given to Packaging, Food and Beverage, Pharmaceutical, Chemical, Logistics and Warehousing, Plastics and Automotive industries.

The vacuum division, which is currently the main driving force of Piab was founded in 1968,

that laid the groundwork for a global presence with a diverse range of vacuum products. Through continuous innovation, Piab expanded its product line, embracing ergonomic handling, robotic gripping solutions, and acquisitions. With over 65 patents, Piab leads the industry, driven by a commitment to research, development, and providing cutting-edge solutions for automated handling. Piab was sold in 2006, and they retain a familial spirit and entrepreneurial ethos under the ownership of Patricia Industries, part of Investor AB, always guided by its inner compass. For Piab to stay ahead of the competition and not fall behind, they have asked themselves how they can create value from the rise of AI.

Piab has taken a deep dive into exploring and providing new internal capabilities to revolutionise internal stakeholder and external customer interactions and experiences. Through the advancements of AI, Piab has developed a Chatbot, named Pixie, which is an assistant available for all our employees, and she can help you with everything – product pitches, translations, researching data, writing emails, responding to emails, taking notes in meetings, and much more.

Since the implementation of Pixie, the adoption of the tool has grown steadily and has been positive. However, the usage still remains moderate to low and scattered between different departments, thus a business need for it to increase. The problem then also becomes, how can Piab Group drive adoption and increase usage of Pixie, in the various departments and divisions, with concrete evidence and insights collected from stakeholders, based on their experience with the tool?

Organisation of Piab group

This stakeholder mapping is utilised to understand the organisational structure of Piab Group (Piab). It is a tool that allows for an understanding of the involved stakeholders and how they interact and impact each other (Newcombe, 2003). This was conducted through a meeting with the Digital Experience Manager at Piab Group. The stakeholder map is high level and it provides an overview of the business itself and the brands it owns. This is important as it allows for the understanding of what a solution should bring to each stakeholder and how a solution can change the relationships between stakeholders. Piab Group is at the top and it has 4 divisions, and each division has brands, as shown in Figure 1.1.1.

Pixie contains specific product database, at the time of our first internal meeting at Piab in March 2024, and documents of Vacuum Automation (VA) and Vacuum Conveying (VC), excluding Coval. The aim is to increase adoption within the three existing divisions, and use these

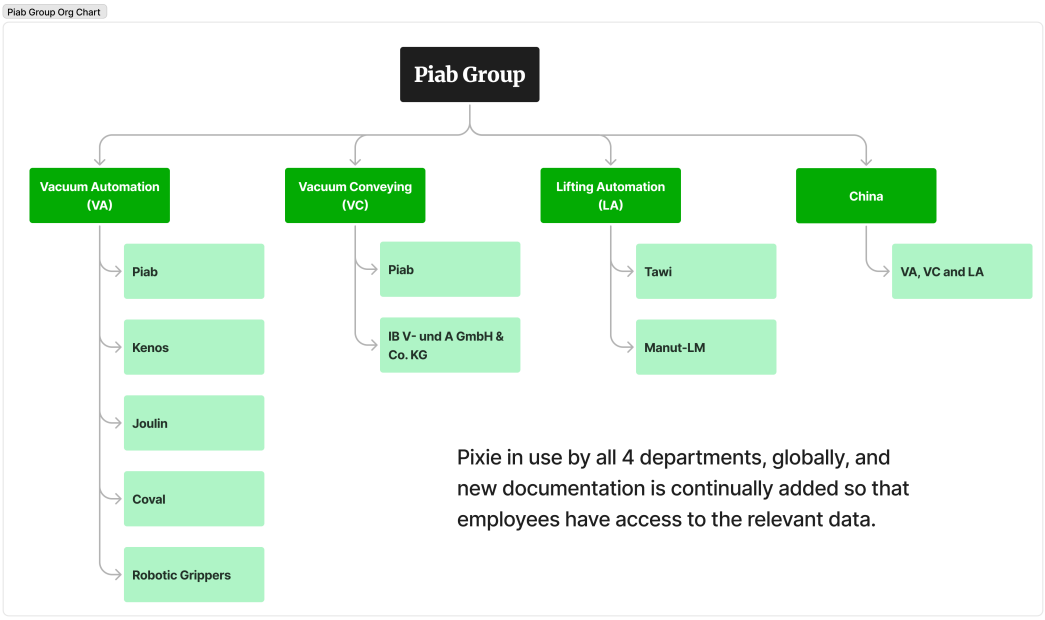


Figure 1.1.1: Piab Group Organisational Chart

successes to expand Pixie adoption and usage in the Lifting Automation (LA).

Pixie - The Personal Assistant AI Chatbot

Pixie, which was developed by Piab, in partnership with Sana Labs, was developed with a easy-to-use interface, with the intention to increase on-boarding satisfaction within the organisation and create one collaborative environment for learning and development. The design of the Chatbot is independent of the Piab website and homepage, as shown in Figure 1.1.2, and can only be accessed by employees.

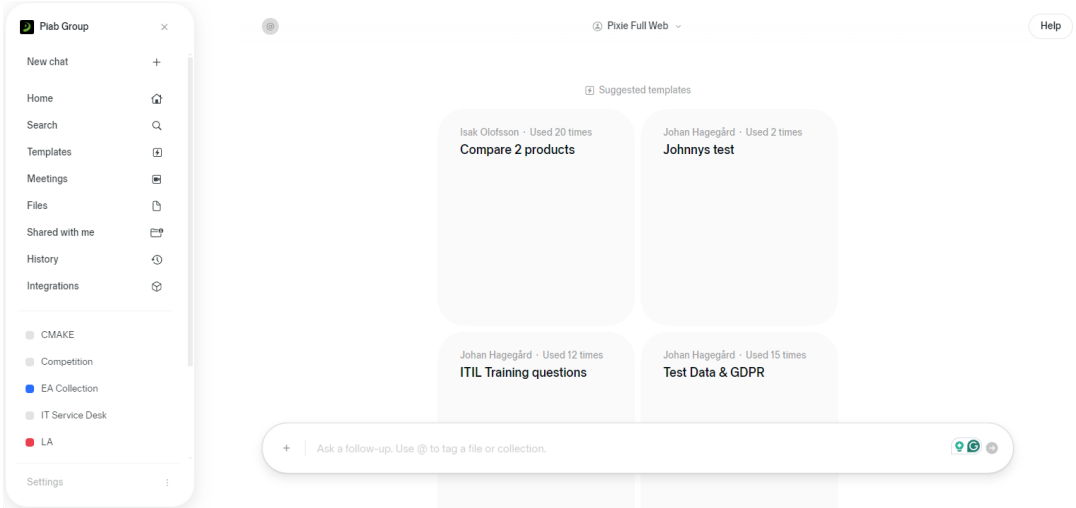


Figure 1.1.2: Pixie Interface

1.2 Purpose

The purpose of this thesis is to explore how artificial intelligence systems, specifically AI chatbots like Pixie, can enhance certain functional areas within an organisation. This study aims to investigate and recommend various approaches that will help increase the adoption and usage of an AI chatbot that could empower organisations to achieve a strategic advantage and foresight in their workplace. It will serve as a roadmap designed to guide all stakeholders involved, old and new to identify factors that influence the adoption and usage of AI chatbots in an enterprise, company, or office setup. This research will investigate the factors influencing the adoption and utilisation of AI chatbots in enterprise settings, examining user characteristics, usage patterns, and the effectiveness of these tools in operational contexts. Research is usually answered with a Research Question (RQ) or multiple RQ's.

The primary focus areas of this thesis are the improvement of sales and operational effectiveness through AI as an assistant. Creating the adoption plan will be crucial to this since it provides a structured approach, mitigates potential risk, and ensures a smooth transition. Through the adoption framework process, this study will explore various outcomes, including enhanced communication and awareness, an enriched database through increasing the data captured, boosted customer engagement and trust, and increased lead generation. This study will aim to fulfil the above purposes through the following RQ's:

RQ 1: What drives the need for AI chatbots in organisational settings and what key features of AI chatbots enhance their effectiveness and value within an organisation?

RQ 2: Which determinants affect the successful adoption and utilisation of AI chatbots in organisational settings and how can organisations overcome these challenges to maximise the benefits of AI chatbots?

1.3 Goal

The goal of this thesis is to investigate and explore the user experience of Piab's AI Chatbot, Pixie, through various research methods, and extract information that will provide insights to improve adoption and usage of the chatbot. The focus would be to investigate the adoption and usage by key departments within the organisation, 2 of which are Sales and the R&D departments, where the biggest positive impact can be made. In sales, the goal would be to find a way that would help the sales team work more effectively. The study also aims

to answer the research questions that the study is structured around and develop actionable insights and recommendations to enhance adoption and usage. The study aims to deploy a mixed-methods approach. The qualitative component focuses on important departments to get in-depth understanding of the AI chatbot's specific contexts and usage. This is supplemented by a quantitative research phase that gathers a larger perspective from a wider range of users across all departments and examines ideas drawn from the qualitative findings. By taking a thorough approach, the study hopes to give a thorough insight of the state of AI chatbot integration today and provide customised recommendations that could greatly increase the usage and adoption of these technologies within the organisation.

1.4 Benefits, Ethics and Sustainability

1.4.1 Benefits

Organisational benefits

Through research, organisations like Piab and companies in industries closely related to that of Piab would benefit as it would bring in new knowledge about application and usage of AI in their business and organisational context. It would provide a valuable knowledge base and reference for companies that are looking to develop and/or implement and use AI tools, specifically chatbots, that implement RAG models, for their business needs. It would provide guidance and insight into use and application cases, to future AI tool developers for such businesses and also to businesses looking to expand on their knowledge and use of AI tools.

Research community benefits

Since the boom and development of highly capable and valuable RAG and Large Language Models (LLM)-based models, AI tools are very new and recent, and the amount of research that has been done from the perspective of AI tools as a product and their related product development is very limited, and studies on their use and application cases are even fewer. This study would add value in the AI research community and to innovation and product development research. It would provide insights into how AI as a product is being used in business perspective and its future prospects. It can also serve as a valuable reference for future studies in AI-based software tools and products, as well as in research studies similar to the one presented in this report.

1.4.2 Ethics

Privacy, anonymity & consent

This study is highly dependent on data captured through communication with individuals employed by the organisation in the study case - the organisation being Piab Group. Since it is data connected to and captured through individuals, it becomes necessary and essential to follow certain ethical considerations. The foremost consideration being consent. It is important to protect the interests of the participants in a qualitative study while driving the interests of the study forward (Gioia, Corley, and Hamilton, 2013). All throughout the research, a protocol of verbal consent was followed before using any data associated with the company and publishing the name of the company. Additionally, during the interviews in the qualitative research, each interviewee's consent to their provided information being used in the study, was confirmed before the start of the interviews. Also, as a general protocol in the study, it was decided to anonymise all the interviewees basic information connected to them and their employment & role in order to preserve their privacy and interests (Gioia, Corley, and Hamilton, 2013).

AI Risks & Dependency

Since this study promotes and explores usage, application and further integration of AI into businesses and work environments, it also acknowledges the dependency on AI increasing and the potential ethical risks involved. According to Buhmann and Fieseler (2021), inherent hazards include the possibility of biases or faults in the data the AI is trained on and the unanticipated negative effects of AI decisions. Significant ethical questions are brought up by this reliance on AI, especially in relation to privacy, and the effects on labour needs and markets. One other concern discussed was the opacity of AI systems—that is, the absence of accountability, transparency, and explain-ability in these systems' functioning. Because

of its opacity, stakeholders may find it difficult to effectively comprehend and assess AI technology, which could jeopardise the circumstances required for responsible innovation and a participation and adoption of AI.

1.4.3 Sustainability

Social context & Inclusive development

Given that the study explores the use and adoption of AI in an enterprise setting for use by internal employees, it becomes important to understand how the users are affected and how to engage them efficiently. This is essential as Kulkov et al. (2023) discusses in their research that in order for organisations to guarantee that AI plans are inclusive and advantageous for all stakeholders, they must successfully engage various groups connected to implementing, developing and using the AI tool. This study aims to involve all groups connected to the AI tool in focus in this research in order to ensure an inclusive development.

Operational Efficiency

Kulkov et al. (2023) discussed in their paper that organisations could enable innovative applications of AI to optimise operational efficiencies that indirectly or directly lead to reductions in energy and resource consumption. Part of this study aims to understand how the subject organisation and its employees use its AI tool and how it contributes towards improving the internal operational efficiency of employees and its contribution towards overall productivity.

1.5 Methodology

In the study of improving Pixie's use and adoption at Piab, combining qualitative and quantitative research methodologies provides a comprehensive understanding of user engagement and satisfaction. This dual approach is crucial for creating effective strategies.

The qualitative portion investigates the unique experiences, perspectives, and expectations of Pixie users and non-users through focus groups and interviews. This method captures detailed narratives about how people use Pixie, the features they value, the challenges they face, and desired enhancements. Köhler, Smith, and Bhakoo (2022) highlights that qualitative research generates ideas based on firsthand feedback, identifies nuances often missed by quantitative data, and addresses specific user needs. However, qualitative research has limitations, such as

scope and susceptibility to bias. Selection and sampling biases may distort results if participants do not accurately represent the entire user community (Šimundić, 2013).

The quantitative method validates and supports the qualitative findings. A survey will be distributed across the organisation to test hypotheses generated from the qualitative data. This approach aims to measure the prevalence of opinions, preferences, and issues identified in the qualitative phase, using a more representative sample to minimise biases. Köhler, Smith, and Bhakoo (2022) notes that triangulation, using multiple methods to study the same phenomenon, leads to more reliable results. The quantitative method provides broader generalisation and statistical validation.

Combining qualitative and quantitative research allows the study to benefit from in-depth user narratives and broad data from a larger user base (Köhler, Smith, and Bhakoo, 2022). This mixed-methods approach enhances the validity and reliability of the results, ensuring that recommendations for improving Pixie's adoption and use are well-founded and relevant to all Piab users and similar organisations. This process guarantees that the strategies developed are based on real data and user experiences.

1.6 Stakeholders

The various stakeholders in the project are noted below:

1. Jennie Björk - Thesis Supervisor (PhD, Associate Professor) from KTH
2. John Hagegård - Thesis Supervisor (Digital Experience Manager) from Piab Group
3. Praneet Kala - Thesis Student
4. Sanjay Rajasekaran - Thesis Student

1.7 Delimitation's

For the time-being, the internal adoption and usage of Pixie will be investigated. The study regarding the external application of the chatbot, outside of Piab, facing the customers and suppliers will be limited or completely omitted due to the available time at hand. We have been made aware of an upcoming experiment with Pixie being used by some of Piab's suppliers and customers to see if it improves sales. This was not considered during the exploration phase due to time constraints. Instead, the focus was on the results from the research methods, primarily

analysing data from the current internal usage of the AI chatbot.

The focus will be solely on the business aspects and how the organisation can gain value through increased adoption and usage, without investigating the technological improvements and solutions that would enable improved adoption and usage of the tool. However, a basic literature study and understanding of AI, chatbots, and the back-end algorithm (RAG models) will be explored to understand how it all fits together.

1.8 Outline

The report on enhancing the use and adoption of AI chatbots, with a specific focus on Pixie at Piab, is structured to offer analysis and practical recommendations to industries and research community, on application of AI chatbots in an enterprise/organisational setting, for internal use. It begins with a Literature Review, where existing research on AI, AI chatbots, RAG models, and their business applications, as well as studies that examine their usage and potential improvements is discussed. This section lays the theoretical reference for the study. The Methodology section describes, in detail, the approach adopted for this study, including both qualitative and quantitative methods used to collect data from users of Pixie at Piab, focusing on interviews and surveys to capture comprehensive user experiences and opinions. The Results section outlines what was discovered from these data collection efforts, highlighting how Pixie is currently being utilised and identifying usage trends and patterns. In the Discussion and Analysis section, a deeper look is taken at these findings, comparing them to the initial hypotheses from the qualitative study and relating them back to the research questions of the study. The limitations of our study and potential areas for future research that could extend our work are also discussed.

Literature Review

This chapter aims to provide an overview of the technological trends and advancements as well as the business case for using Chatbots. The chapter begins by presenting a broad overview of AI, how it is used with Chatbots, and a high-level overview of the algorithm behind Pixie. The literature then continues to describe how this technology can be used within businesses to create value within teams, making them more effective. It also discusses the gaps in the current literature and provides a base for the formulation of the proposed theoretical framework.

2.1 Artificial Intelligence

Since the term AI was first coined in 1955 (McCarthy, Minsky, and Shannon, 1955), there has been a rapid development of AI technology in information communications, manufacturing, and related product technology (Li et al., 2017). A widely accepted description of AI defines it as a technology that enables machines to replicate complex human capabilities (Sheikh, Prins, and Schrijvers, 2023). Artificial intelligence, particularly Machine-Learning (ML), is regarded as the foremost general-purpose technology of our time, as it highlights the machine's capacity to continually enhance its performance autonomously, without requiring explicit instructions from humans for every task (Brynjolfsson and McAfee, 2017). It can be seen in Figure 2.1.1 that AI consists of several other faculties, which makes it an extensive and broad topic. It has also grown and developed over the years, especially in the field of Machine learning, Natural language processing and cognitive computing.

Sheikh, Prins, and Schrijvers (2023) emphasises the importance of understanding the differences between human and AI to optimise interactions and engagements. By acknowledging the distinct cognitive abilities and constraints of each system, individuals can strategically allocate tasks, leverage the strengths of AI systems, and complement human decision-making processes with data-driven insights. This understanding is vital for creating a collaborative framework that maximises the distinct qualities of both humans and AI systems, leading to more efficient and effective outcomes. The framework aims to cultivate intelligence awareness to better comprehend, investigate, and address the diverse possibilities and challenges of machine intelligence. By developing new, targeted, and easily configurable training forms and learning environments, the framework seeks to equip individuals with the knowledge and skills necessary to navigate the complexities of human-AI interactions

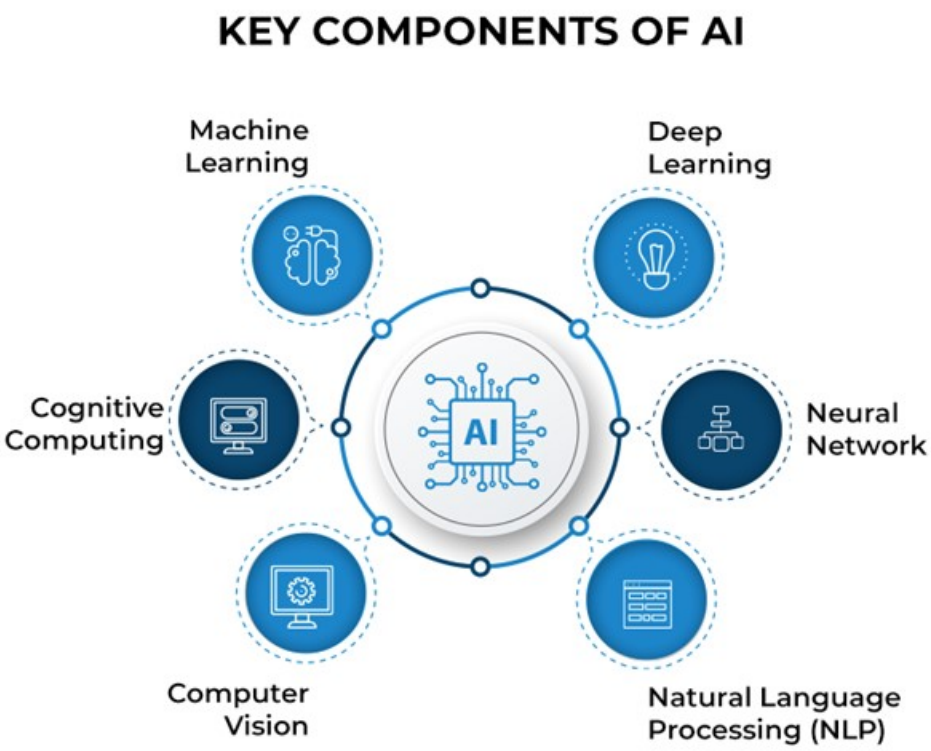


Figure 2.1.1: What is AI (Kanade, 2022)

effectively (Sheikh, Prins, and Schrijvers, 2023). This framework allows businesses to grow and evolve with AI by continuously collecting data and learning how the organisation can adapt to technological advancements.

The investigation performed by Di Vaio et al. (2020) provides a quantitative overview of how AI plays a crucial role in the construction of sustainable business models by enabling companies to integrate sustainability principles into their operations and decision-making processes. They discuss many key ways in which AI can positively contribute to supporting businesses. Enhancing operational efficiency and productivity that can lead to cost savings and a more sustainable use of resources. AI-powered tools, such as Chatbots and personalised recommendations, can enhance customer engagement and promote sustainable behaviours. Other points they touch on include fast data analysis and decision-making that quickly and accurately provide valuable insights for informed decision-making, ensuring that companies foster an environment of innovation in product development and risk management strategies related to sustainability issues (Di Vaio et al., 2020).

Davenport and Ronanki (2018) discuss how AI can support three business needs through the lens of business capabilities. Automating tasks and processes, especially administrative and business processes, was quick and effective and the least expensive. Performing a thorough data

analysis on data gathered to gain cognitive insights was also valuable. This is where the power of ML comes into play. According to their study, cognitive insight applications are commonly employed to enhance the efficiency of tasks exclusive to machines, like programmatic ad buying, which entails rapid data processing and automation beyond human capabilities. The last need is cognitive engagement with clients and employees. It was the least common type of engagement to use chatbots, intelligent agents and ML. This could be due to the fear that employees will lose their jobs, which is true to some extent, since they believe cognitive systems perform tasks, not entire jobs (Davenport and Ronanki, 2018).

From an organisational perspective, Piab Group has a long-term strategy to stay ahead with technology and innovative ideas, aiming to become an AI-first organisation and a thought leader in this domain, being at the forefront or making this a success. They are focusing on integrating chatbots within the organisation to act as personal assistants, seamlessly fitting into employees' daily routines.

2.2 Chatbots

Chatbots, powered by AI, engage in realistic conversations with people, gaining popularity for their focus on natural language and user interfaces in messaging apps (Adamopoulou and Moussiades, 2020; Ali et al., 2024). AI chatbots are designed to undertake routine and low-level cognitive tasks, with some of them serving the additional role of virtual assistants through human interaction with computer systems (Adamopoulou and Moussiades, 2020; Bhoir, Patil, and Mogul, 2022). Designing a meaningful chatbot experience requires a thorough understanding of user expectations for this technology and identification of opportunities that align with diverse needs (Steinbauer, Kern, and Kröll, 2019). It is also quite an advanced technology, and they have the potential to create a huge positive impact in organisations. Hsu et al. (2023) conducted a study on Chatbots vs mobile apps and the results that show that Chatbots excel in one-attribute, information-light, and group-buying tasks, while traditional apps are better suited for multi-attribute, information-intensive, and single-buying scenarios.

In the field of natural language processing (Natural Language Processing (NLP)), chatbots can be quite an advanced technology (Majumder and Mondal, 2021). NLP utilizes computational techniques to learn and understand language by automating the analysis of linguistic structures (Hirschberg and Manning, 2015). This involves the development of fundamental technologies, including machine translation, speech recognition, and speech synthesis. It is a branch of computer science, employs computational methods to comprehend, learn, and produce human

language content. Computational linguistic systems have diverse applications, aiding human-human communication in areas like Machine Translation (MT) and improving human-machine interaction with conversational agents. Moreover, they contribute to both human and machine knowledge by analysing and learning from the extensive pool of human language content.

Another important tool that is used when developing Chatbots is the presence of LLM, which are essentially artificial neural networks. Large Language Models LLM's are sophisticated AI systems capable of understanding and generating human-like text. Trained on vast amounts of data using deep learning techniques, these models are built on transformer architecture. This architecture allows them to process information efficiently and perform various natural language tasks, such as generating text, classifying information, and translating languages (IBM, 2024).

With all of this technology in place, Piab Group is among the few companies that have successfully implemented a Chatbot model that accesses and augments information from their own database to provide unique feedback to users. This next-level chatbot technology can access proprietary information specific to the organisation and its brand, retrieving data from secure databases and augmenting it with LLM to generate accurate responses. Companies are increasingly seeking ways to leverage this cost-effective technology to optimise employee queries and outputs.

2.3 Retrieval-Augmented Generation Models

RAG models is a recently emerged and promising solution that makes use of external databases. In the context of organisations using the RAG models, it means using their own data that is stored in external databases and not from open-source sources. RAG is a cutting-edge approach that optimises the output of LLM by integrating external data retrieval into the generative process (AWS, 2024; Chen et al., 2023; Gao et al., 2024). This innovative method mitigates hallucination in LLM, enhancing their capacity to deliver accurate and relevant responses by referencing authoritative knowledge bases beyond their training data sources (Chen et al., 2023).

The basis of a RAG model is having a specific knowledge base that is internal to the entity or organisation. A user asks a question and the chatbot will lookup and retrieve information from the specific knowledge base, augment this and send it to an LLM and then generate an answer, as shown in Figure 2.3.1. It is a very high level overview and there can be lots of layers beneath

each step, but the flow is the same. It also matters what LLM (Application Programming Interface (API)) to utilise, it's essential to factor in considerations such as privacy concerns and cost limitations and fortunately, several viable options are already available, including OpenAI's GPT, Meta's LLaMA, and Google's PaLM.

This method represents a significant advancement in AI, allowing organisations to leverage their own data securely and efficiently to enhance the accuracy and relevance of chatbot responses.

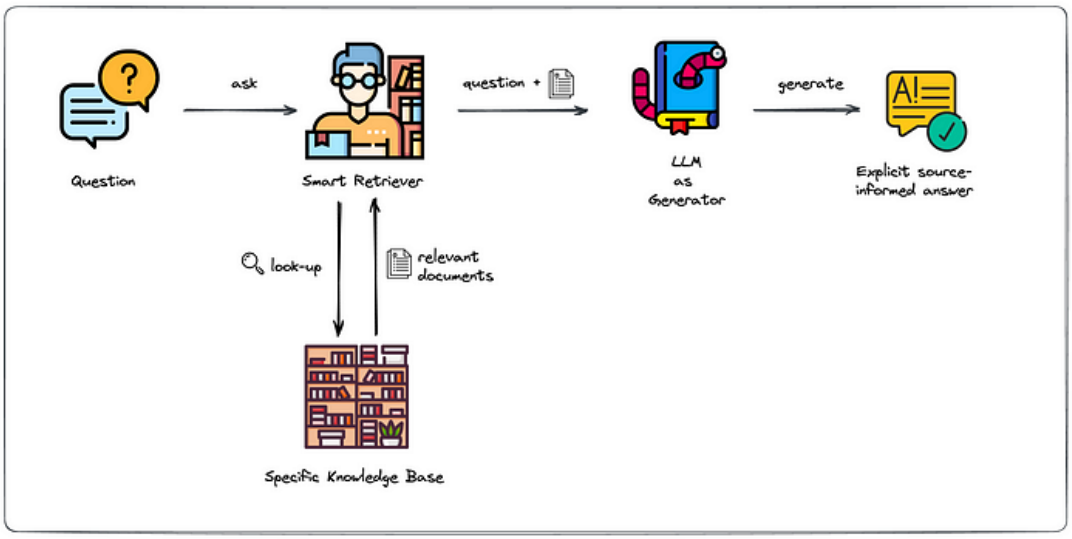


Figure 2.3.1: Overview of RAG Model(De Koninck, 2023)

2.4 Business Case

AI has long been a part of our life through products and tools. Research conducted by Soni et al. (2020) concluded that the progressive growth and application of AI driven systems is attributed to two main factors: availability of big data and fast processing units. They also mention that AI based products create pathways to explore improved opportunities in several avenues including business intelligence, marketing and sales. They say that the developed AI algorithms help improve performance factors like time, cost and efficiency. Keeping Piab's AI chatbot in mind and the fact that Piab can generate big data from the products being used by its customers, this presents a huge opportunity for Piab to develop a new type of offering to its customers to help improve their performance through their products with the use of automated big data analysis and recommendation that AI based LLM is capable of. For instance, Piab could implement AI chatbots to provide instant product recommendations, predictive maintenance alerts, and real-time customer support, thereby enhancing customer satisfaction and operational

efficiency.

Lee et al. (2019) says that having sufficient, high-quality data is paramount for success, emphasising the crucial role of data acquisition and infrastructure in business transformation, and this is shown by their virtuous cycle of AI as shown in Figure 2.4.1. Data is continuously being collected, and as the adoption of the tool grows, so will usage, increasing the data, therefore providing more information to create informed decisions. They can also use this tool as an efficient Enterprise Resource Planning (ERP) and Enterprise Resource Management (ERM) systems within the company which can be used by all departments within the company for their specific suited purpose to improve efficiency and reduce cost and time spent on data analysis, research and daily routine tasks. For example, Piab's sales team could use AI chatbots to quickly access customer data and sales insights, while the HR department could automate recruitment and training processes.

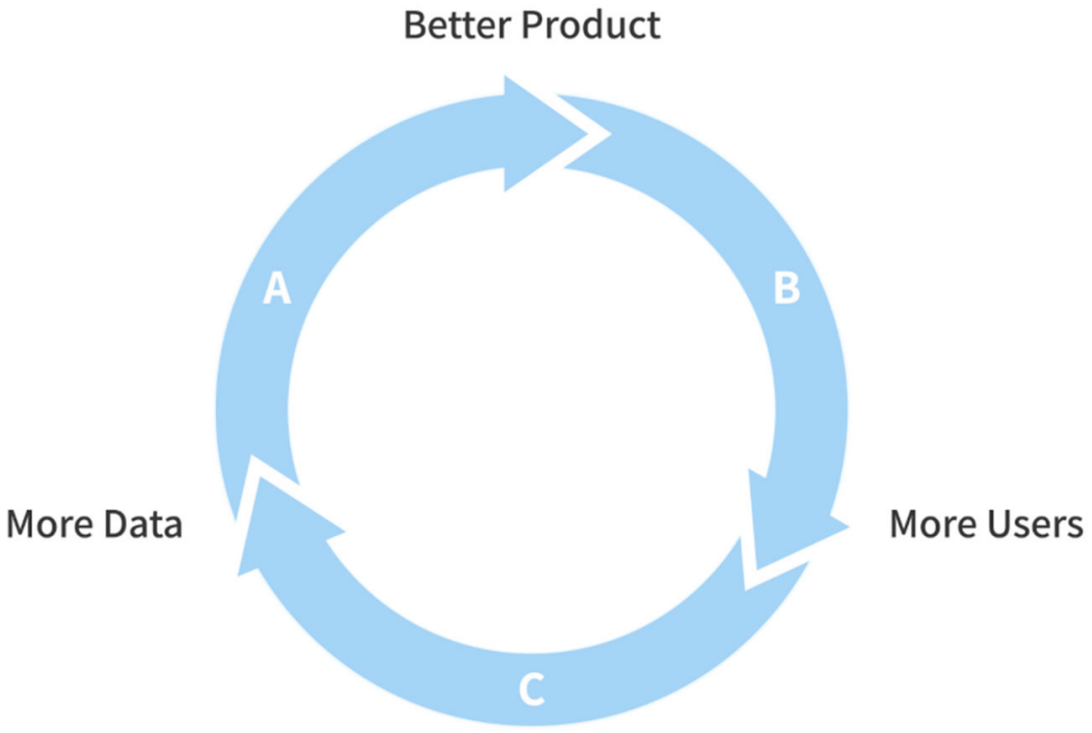


Figure 2.4.1: Virtuous cycle of AI (Lee et al., 2019)

In research done by Kaczorowska-Spychalska (2019), it highlights the fact that more than 67% of consumers used chatbots for customer service last year, indicating the growing acceptability and application of chatbots in customer assistance. These consumers include individuals seeking customer support, product recommendations, or information about services and brands. It goes on to say that by 2022, chatbots are expected to save $8 billion in company expenses, making them an invaluable tool for organisations and possibly yielding significant profits. This

ability stems from its capability to improve productivity, reduce response times, improve overall customer service, and manage routine queries. This leads to a reduction in customer support costs and operational expenses for businesses. It also mentions how chatbots are increasingly being used in internal marketing processes, such as recruiting, training, and offering assistance for specific marketing goals, tools, and strategies. Additionally, the analysis demonstrates that the sectors with the most projected demand for chatbot solutions are those like e-commerce, insurance, financial advising, and telecommunications that now have a high degree of user acceptability. This indicates that businesses across several industries are recognising the benefits of integrating chatbots into their operations to enhance customer experiences and streamline workflows. For Piab, this means there is a clear business case for implementing AI chatbots to not only improve customer interactions but also streamline internal processes.

Kaczorowska-Spychalska (2019) also mentions that the full business potential of chatbots or AI-based systems interacting and cooperating with each other is not fully explored. This concept involves creating networks where chatbots or AI systems can perform required tasks while interacting and collaborating with other systems/chatbots exclusively. This opens new possibilities for Piab to create a network of chatbots that can handle complex tasks, such as coordinating between different departments or managing end-to-end customer service processes.

In research done by Nagarhalli, Vaze, and Rana (2020), they say that chatbots can reduce the workload at higher management levels and in enterprises in general by up to 70%. This is possible due to automated routine tasks and give instant, accurate and specific responses to searches which in turn improve operating efficiency of employees in an enterprise. The research also says that processing of unstructured data provided in the form of text data as knowledge has not been explored enough.

2.5 Improving usage of AI chatbots

Here, exploration of research works that are associated with aspects related to use of AI chatbots by users in a workplace or as customers, is done. This is done in order to understand and narrow down on the characteristics, capabilities and expectations from an AI chatbot that makes users want to adopt and use it more. Findings from this section would also help in framing the methodology for the research being conducted in this paper.

The paper written by Gkinko and Elbanna (2022) was quite informative, where they conducted

a research involving an case study regarding the emotional experience of employees using an AI chatbot and how those emotions impact and influence its use. A large multinational organisation that had implemented an AI chatbot in its Information Technology (IT) support functions, was considered for the case study setting. The chatbot was intended to help employees resolve common IT problems and other Frequently asked questions (FAQ). The case study and data collection was conducted over a period of 12 months and the following methods were used for the process:

1. Interviews

   A total 41 unique employees were interviewed to understand their experience and emotions associated with using the chatbot at the workplace. The employees were distributed across different levels of roles from senior executives, middle managers and junior staff. 3 rounds of interviews were done. In the first round, the product manager, project manager and two software developers, all associated with the implementation of the chatbot were interviewed. This was done to understand the features, implementation decisions, technical aspects, organisational objectives, project plan and future vision regarding the chatbot. In the second and third round, the users of the chatbot were interviewed and observed to understand their experiences and emotions and how it influenced their use of the chatbot. The interviews were semi structured and conversational in nature. Most of the interviews were conducted online and the researchers felt the online interviews were more effective, avoided typical office interruptions and lengthy introductions and closings.

2. AI chatbot project document reviews

   The documents that were reviewed by the authors included project documents, internal corporate newsletters, organisational announcements and emails, and internal links. This was done to better understand the context of the chatbot's implementation, introduction, adoption, development and technical features.

3. Observation of use

   The use of the tool was additionally observed by the authors during the interview by requesting the interview candidates to share their screens when using the chatbot and share screenshots of the problems or occurrence that they associate their emotions and experience with.

The authors identified characteristics of the AI chatbot that triggered emotions that encouraged users to use the tool more. Form & function (features of the chatbot that define its form and

function like the conversational interface and its presence and connectivity in the company network etc.), machine learning capability (chatbot's ability to learn and improve from user interactions and improve responses universally when corrected), automation & process design (the automated functions the chatbot is capable of and its effectiveness) and user interface design. They also discovered that the users usually experienced emotions ranging from connection, contentment, amusement and frustration.

While the growing trend for the need of application of AI tools can be observed, there still wasn't specific attention directed towards the need for an exclusive LLM based AI chatbot in the workplace. It was also observed during the literature review phase that, there were no significant case studies or research done with respect to application/implementation study of AI chatbots in the workplace. There was also a lack of references observed when it came to guiding materials and frameworks for implementing such tools and how companies in various industries are incorporating AI, specifically LLM chatbots in their businesses and workplace and addressing of faced challenges. Understanding the determinants and factors that drive usage and adoption of AI chatbots and developing strategies to address them is crucial for organisations aiming to maximise the benefits of AI chatbots. To understand them, application case studies and observational studies are required to identify patterns and trends. This leads to the central research questions of this study, which seek to explore what drives the need for, use and adoption of AI chatbots in organisational settings, what key features enhance their effectiveness, and how organisations can overcome the challenges associated with their adoption and utilisation.

Method

This chapter provides an outline of the research design, detailing the selected approach, methodology, presentation of data collection methods, and data analysis techniques employed in this study. It also includes a discussion on the procedures implemented to ensure the credibility of the findings, encompassing strategies for establishing validity and reliability.

3.1 Research Design

The research design of this project involves collecting data from Piab Group employees through interviews, a survey, and the AI chatbot system, followed by an in-depth analysis of the results. This comprehensive analysis aims to address the central research questions of the study, with findings presented in the report through detailed discussions supplemented by visual aids. A mixed-method approach was employed, consisting of both qualitative and quantitative studies. This approach seeks to understand how the adoption and usage of Pixie can be increased and what factors influence and drive employees to use the tool. Emphasis will be placed on increasing the usage of the chatbot. This factor can be used to identify performance bottlenecks, understand usage patterns, quantify productivity enhancements, and assess the performance of various features or services. These insights can impact product development, marketing strategies, and investment priorities within the organisation.

In order to make informed decisions based on business insights, it is essential to collect accurate data that can be systematically analysed and processed, thereby providing answers to relevant business questions. This approach ensures that strategic decisions are supported by empirical evidence, enhancing the reliability and effectiveness of business operations. Collecting data from various streams, as mentioned above, allows for the assessment of general situations, rather than just specific cases. By consistently gathering, tracking, and analyzing data, progress or lack thereof can be measured over time. The research method involved a combination of data collection methods, two of which are described by W. J. Creswell and J. D. Creswell (2018) as: an inductive approach (qualitative) to gather subjective experiences and a deductive approach (quantitative) to validate, invalidate, or provide support for established theories. Qualitative research fundamentally employs textual data, which is collected and analysed through various methods, and quantitative research relies on numerical data, which is examined using statistical

methods (Clarke and Braun, 2013). Utilising two different frameworks highlights the different approaches and tools that are used to explore, interpret and understand the data.

Planning the research ensured a methodical and organised approach to the problem. The plan entailed a systematic division of the research study into four distinct phases, each designed to address specific aspects of the research objectives. These phases are comprehensively discussed in the following sections, providing detailed insights into their methodologies and anticipated outcomes.

3.2 Phase 1 - Data Sorting and Interviewee Selection Process

Phase 1 involves the preparatory work for the qualitative research, as illustrated in Figure 3.2.1. To understand the user's perspective—critical for the qualitative aspect of the research—a strategic selection and analysis of Pixie users was required. This analysis needed to identify who is currently using Pixie, the frequency of use, and the number of chat messages sent. Here, 'chat messages' refer to each prompt or question submitted via the Graphical User Interface (GUI).

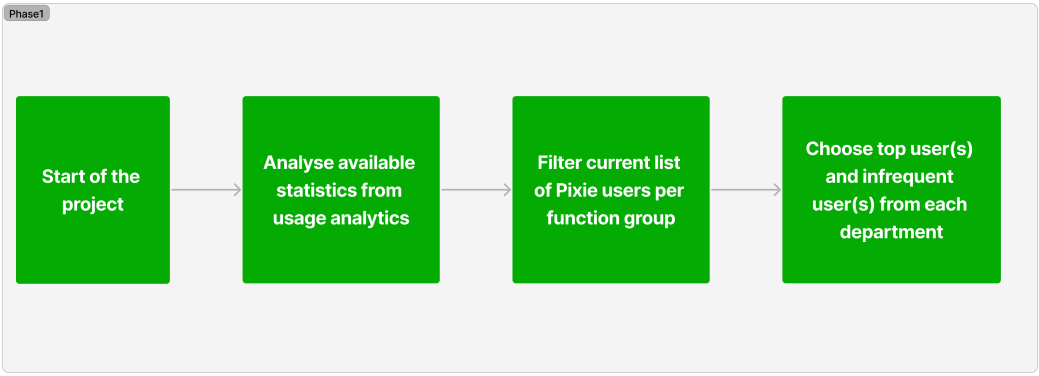


Figure 3.2.1: Phase 1 - Data sorting and Interviewee selection

Piab receives weekly usage reports, which include user emails and the number of chat messages sent per month. Sana Labs, the provider of the Pixie platform, facilitates access to the platform and integrates it with Piab's IT infrastructure, including Microsoft (MS) Teams, Outlook, and SharePoint. All usage statistics gathered from Pixie are shared with specific members of the Global Digital and Information Systems (GDIS) team through weekly Excel reports. For this research, the relevant data pertained to the number of chat messages each user sends per month, which helps gauge utilisation trends over time, including any increases or decreases. Additionally, a more detailed report was made available, showing the total number of chat messages sent per week, with date/time stamps for more accurate readings.

In order to visualise the data, Piab have developed and implemented a Power BI dashboard that displays several metrics from the weekly usage reports of Pixie. Power BI dashboards serve as a comprehensive visualisation tool within the organisation, enabling data collection from various sources, transformation of data for analysis, and the creation of interactive visuals for detailed examination. It facilitates collaborative decision-making through data sharing and offer real-time updates for operational monitoring. The AI capabilities of Power BI also enhance the discovery of patterns and trends, aiding in the uncovering of valuable insights. This all contributes to promoting data-driven decisions and keeping the organisation informed about crucial metrics and trends. A newly created Power BI dashboard was developed to identify trends and correlate them with internal or external events affecting Piab.

The company employee data was also compiled into an Full-Time Employment (FTE) report in an Excel spreadsheet. Piab maintains a comprehensive list of all employees, including details such as user names, emails, country, function, division, and more. This list served as a crucial data source for the study, enabling the cross-referencing of employees with the data provided in the weekly reports. However, not all the information provided was necessary for analysis and some data needed to be filtered out. Preprocessing of the data was necessary because Piab has two email domains, and each employee has four email addresses (alias email addresses). The data included multiple email accounts for the same individuals, necessitating cross-referencing with the FTE report from Piab to eliminate duplicates and ensure that no redundant data was included. It was later noticed that some of data from the weekly reports consisted of persons that were not in the FTE report, either being external consultants, internship students, short-term employees or ex-employees. (Discussed in section 5)

The data from the usage metrics report and Piabs' FTE report were imported into Power BI for further analysis. Certain columns were adjusted, and the table containing the usage data

was pivoted to facilitate easier handling. A many-to-one relationship between the two tables were established, as illustrated in Figure 3.2.2. This relationship enabled the comparison of chat messages with various other metrics, which was subsequently used to analyse and discuss the results from both qualitative and quantitative perspectives. This approach was instrumental in identifying trends, creating forecasts, illustrating themes, and effectively communicating complex research findings.

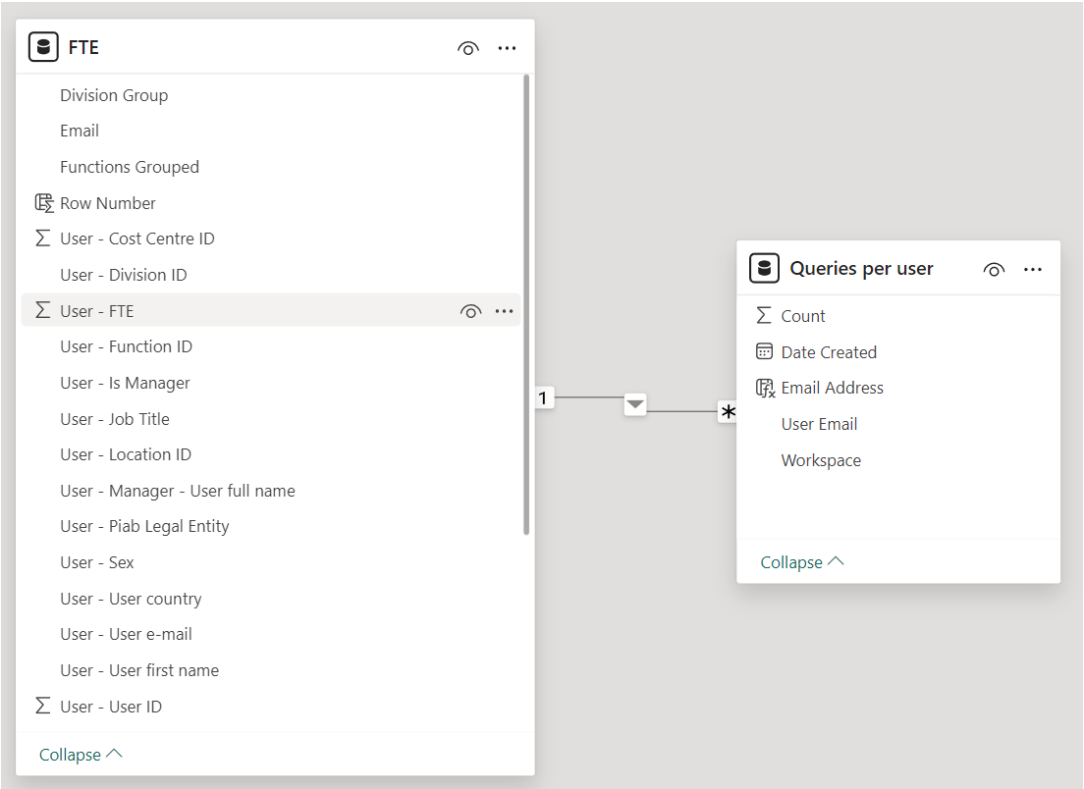


Figure 3.2.2: Power BI Relationship connection

To facilitate the identification of employees, the data was categorised into function groups. Functions, within the context of Piab, refer to the various departments within the organisation, such as sales, procurement, engineering, IT, and Research & Development (R&D), to name a few. These groupings allowed for a more structured analysis of usage patterns and trends. The metrics that are important and relevant to this research were the *number of chat messages* and the *number of existing users*. These measures helped identify which function groups to target for our interviews. This data was also used to identify trends and patterns in usage and adoption rates across different regions, countries, and specific teams. The selection of top and infrequent users was made from the processed data within each functional group. Narrowing down the list of employees was performed on the two pools of users in each functional group. Several assumptions were made, including that the employee would:

1. Have their work and daily tasks dependent on the use of a computer,and hence have definite and prolonged access to a computer and hence to Pixie.
2. Benefit from using Pixie in their work in terms of efficiency and productivity and result in improving the business's overall productivity and performance.
3. Were not aware of Pixie and have never used it before.

Later, it was discovered that information from non-users were not part of the list. Although they were eventually considered, they either did not respond to the interview invites or did not wish to partake. Timing became a problem at that point, which is discussed in section 3.3.2.

Furthermore, the process of narrowing down and selecting employees from specific function groups was carried out. Due to time limitations, the number of interviewees had to be restricted. Another limitation was the large staff count, which made it impossible to interview everyone. A strategic decision was made in collaboration with Piab to interview specific individuals from different function groups and divisions. The function groups chosen were:

1. Inside Sales
2. Outdoor Sales
3. R&D Team
4. Engineering Team
5. Operations Team
6. Marketing and Communications
7. Management Team
8. Business Development

The decision to choose these groups varied, all aimed at increasing sales, productivity, efficiency, and the usage of the chatbot tool. Initially, five candidates were selected from each department for interviews. In addition to the initial list, a few more individuals were identified internally as likely to provide valuable insights and were subsequently added. A more detailed description of the data collection can be found in section 3.2.1. The users selected were chosen at a specific time, and that data was used for the duration of the interview selection process. [1]. Since weekly updates of usage statistics were provided and the interviews were planned to start from April 1st onwards, new "super-users" emerged during this period who surpassed the initially selected users. These new super-users were kept in reserve in case of cancellations or declines to participate. Users who were previously unaware of Pixie or had not used it might have updated usage metrics, and these updated statistics could be reflected in newer results.

[1]For the purpose of this section, and due to time constraints, the data that was used in the method was dated 18th March 2024 and has all records and metrics prior to that date. The data collected and analysed after this date is only used in the sections 4, 5 and 6, and has been shared with Piab for internal use.

This brought the list of participants to 38 individual employees from different functional groups and regions.

3.2.1 User Data Analysis/Data Mining

The data collection and analysis conducted in the research method involved processing data from Excel sheets. The sample data files, part of Piab's regular weekly data, were provided by Sana Labs. The initial dashboard, designed to share insights with General Management (GM), was incomplete and required minor adjustments. New graphs and charts were created from the data specifically for this research project. Additional data processing tasks were performed to identify trends, patterns, and usage statistics in the visualisations. It is important to note that some data files were corrupted, but these issues were resolved in subsequent weeks, allowing for accurate monthly updates.

To supplement the data analysis and gain a deeper understanding of user behaviour, the provided data was used to identify individuals within specific divisions and function groups. Additionally, details were extracted on users who sent the most and the least messages over several months, from the tool's launch in August 2023 until March 11, 2024. Piab has several function groups, and the data was filtered for each group to highlight a few top users and a few infrequent users, as illustrated in Figure 3.2.3.

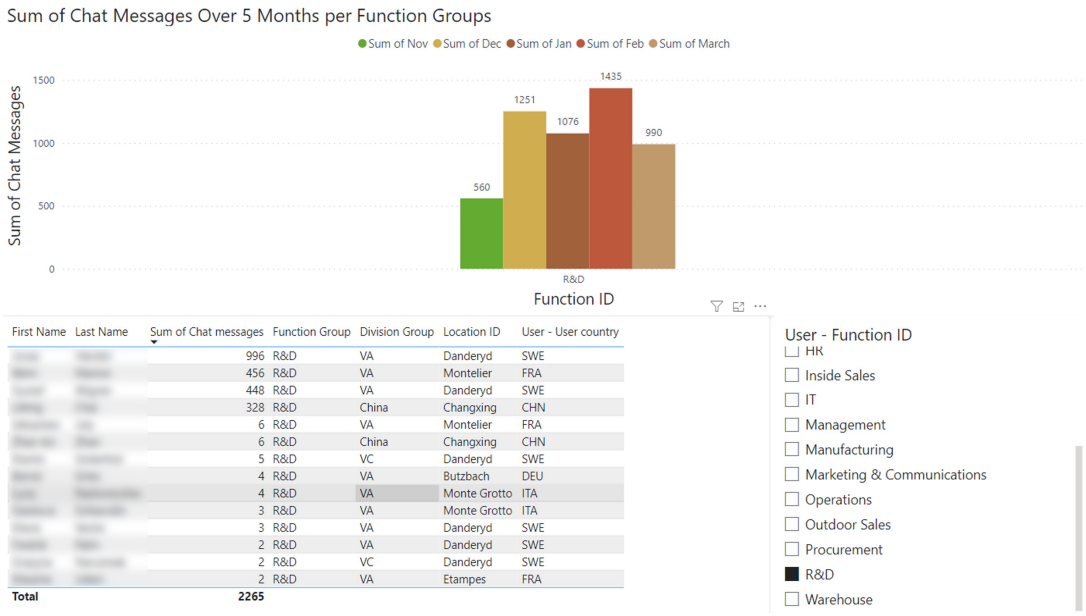


Figure 3.2.3: List of users in R&D

Countries were also considered, ensuring a comprehensive understanding of usage patterns across different regions. The column chart in the figure has been filtered to display only the

R&D team, providing a visualisation of the overall statistics per function group. The function group is selected from the ”slicer” selection box located at the bottom right of the image. The accompanying table includes several columns of information that were instrumental in determining which individuals were selected for interviews.

3.3 Phase 2 - Data collection through interviews

The second method of data collection in this study involved conducting interviews to gain insights from stakeholders regarding the project and establish Piab’s needs. This facilitated the performance of thematic analysis, enabling the recognition, scrutiny, and comprehension of significant patterns (Attride-Stirling, 2001; Clarke and Braun, 2017). This approach allows an organisation to identify specific pain points, preferences, and challenges encountered by employees, thereby providing valuable qualitative data on the current perception and usage of the chatbot. The interviews were semi-structured, allowing interviewees to discuss the topic more freely and offering an in-depth examination of the subjects under investigation (Bryman and Bell, 2017).

The focus of this phase was to develop an interview guide (as shown in Figure 3.3.1) that would form the foundation of the qualitative study. This involved reviewing available literature on qualitative analysis of AI chatbots in the workplace. A schedule was developed to coordinate and plan the interview process, taking into account the time zones of employees in different countries. Interview invitations were sent to employees in advance to accommodate their schedules.

3.3.1 Interview Guide

The interview guide was designed to facilitate a smooth interview flow for the participating candidates, enabling seamless transitions between questions. It comprised an introductory section, followed by five core thematic sections, and concluded with a closing section. This structured approach also streamlined the information categorisation during the analysis phase, as similar responses could be efficiently grouped. The full interview guide is available in Appendix A.

The interview guide was designed to provide **background information** about the research to the interviewees. A brief overview was also included in the invitation email. The initial section addressed informed consent, anonymity preferences, and recording permission. Gioia,

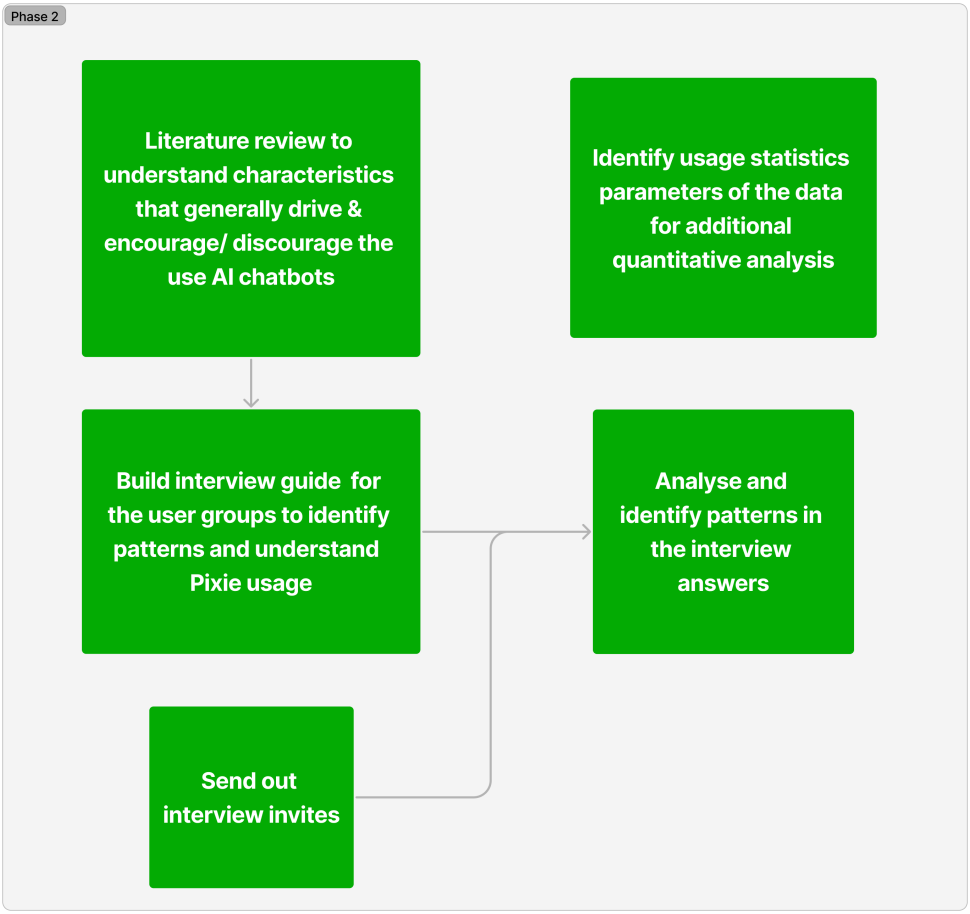


Figure 3.3.1: Phase 2 - Data collection through interviews

Corley, and Hamilton (2013) highlights in their research on qualitative research methods and conducting interviews, it's crucial to balance the researcher's needs with protecting the interests of interviewees, particularly those employed by the focus organisation. Transparency is key, and while complete confidentiality cannot be guaranteed (meaning some aspects of their responses may be used), anonymity is assured by default.

Understanding user adoption and **usage patterns** was a core research objective in the project. A set of **background information** questions were included to gauge the interviewee's level of adoption (user or non-user). These questions explored the interviewee's:

- Awareness of Pixie
- Source of their knowledge about Pixie
- Duration of use (if user) or reasons for non-use
- Initial impressions of the new AI tool

Building on the work of Gkinko and Elbanna (2022), the guide delved into user experience and **usage patterns**. A section focused on **usage patterns** explored frequency of use, typical use cases, and their last experience using Pixie in relation to their job role (open-ended). The user experience section examined satisfaction with Pixie, experiences with Pixie failures (open-ended), and features found most useful (open-ended). To understand Pixie's impact on job roles, open-ended questions explored situations where interviewees perceived a positive or negative impact on their work and productivity.

Finally, a **feedback** section addressed potential new features, views on existing and future functionalities, training needs, and Pixie's current visibility and accessibility. The interview concluded with thanking the participant and explaining how their responses would be used.

3.3.2 Interview Candidates

The interview guide was designed to elicit key insights into user experiences and perceptions of the tool. Due to time constraints (30 minutes per interview), the number of questions was limited, ensuring conciseness and directness to facilitate a larger sample size. Out of the 46 candidates contacted via email, 24 semi-structured interviews were successfully completed. An extract of the information is presented in Table 3.3.1. To maintain the privacy of the interviewees, the names of the employees have been omitted; however, all other data shown in the table is accurate.

It is important to acknowledge limitations in data collection. One interviewee did not provide consent for data use and recording, necessitating the exclusion of their responses. Additionally, the CEO's interview was excluded to mitigate potential bias towards leadership perspectives.

The response rate was not uniform across participants during the interview phase. Thirteen emails received no response, and six individuals declined participation due to various reasons (unwillingness, sick leave, timing constraints, retirement, or lack of Pixie experience). Two participants provided partial responses via email, which will be considered in the analysis.

This resulted in an uneven distribution of participants across departments, as reflected in Table 3.3.2. Due to the lack of *engineering* personnel, data and insights for the *engineering* function will be collected through the survey and presented in the results section. Unsurprisingly, most interviews were conducted with the VA division, the organisation's largest. While candidates

from VC and LA either declined or did not respond, one GM member from VC provided valuable input that has been considered.

3.3.3 Interview Coding

Following the interviews, all transcripts were coded using the thematic analysis method outlined by Attride-Stirling (2001) to facilitate structured data analysis and comparison of findings. A bottom-up approach guided the coding process. Initially, basic themes were identified and then grouped into 12 overarching themes based on their similarities. These overarching themes were further refined into 3 core global themes. To analyse the prevalence and variation of basic themes across interviews, a pivot table was used to count the occurrences of each overarching theme, noting how frequently specific basic themes appeared within each interview. Maintaining consistency in the creation of overarching themes was crucial to prevent an overabundance of themes lacking substantial support from basic themes.

The three global themes that were identified are:

- **User Experience, Technical Challenges, and Future Enhancements**: This theme explores users' direct interactions with the tool, encompassing their experiences (positive and negative), technical challenges encountered, and suggestions for future improvements. It delves into a range of subtopics, from user interface issues to specific technical problems and desired enhancements, highlighting areas for optimisation and user satisfaction (as expressed by interviewees).

- **Impact on Professional Growth and Efficiency**: This theme delves into how the tool influences users' professional lives and working lifestyle. It examines user perspectives on whether the tool enhances or hinders their productivity and professional development. Insights gained from this theme will reveal the effectiveness of the tool in boosting job performance and efficiency, as reported by the interviewees.

- **Responsible Innovation and Trust in Digital Transformation (Policy)**: This broader theme explores user concerns surrounding the ethical implications and trust factors associated with the tool and digital transformation in general. Data privacy, security, and the ethical use of AI are key areas of focus within this theme. Examining these aspects will generate deeper discussions about the organisation's culture, governance practices, and the role of digital transformation in modern workplaces, as reflected in user feedback.

3.4 Phase 3 - Data collection through survey

Phase 3 of this study adopted a quantitative research methodology to comprehensively assess the efficacy and utilisation patterns of Pixie. A structured survey instrument was employed to collect primary data, while secondary data was derived from the analysis of weekly reports. These data sources collectively enabled a thorough examination of chatbot usage frequency, user satisfaction levels, and interaction dynamics. The overarching aim of this phase was to quantify Pixie's influence on organisational processes, discern emerging usage trends, and provide empirical evidence to inform strategic enhancements.

The survey instrument development was guided by the overarching themes and granular subtopics identified in the interview guide as shown in Figure 3.4.1. The survey questions were crafted to elicit detailed respondent perceptions and behaviours related to Pixie. By adopting this approach, the research aimed to generate a robust dataset amenable to rigorous analysis, ultimately informing recommendations designed to optimise the chatbot's functionality and user experience.

Subsequent to data collection, a systematic analysis of survey responses was conducted, incorporating relevant statistical measures. To enhance understanding, survey findings were compared with data from weekly reports. This comparative analysis facilitated the identification of patterns and trends in user interactions with Pixie, informing evidence-based recommendations for chatbot improvement.

3.4.1 Survey

Formulating the survey was a crucial step in data collection for this quantitative study. Building upon insights gained from the interviews, the survey aimed to gather a broader perspective from the organisation. It served as a complementary tool to the interview data, providing quantitative information for analysis. The survey was distributed to all 1230 FTE employees and remained open for fourteen working days. To maximise participation, it was disseminated with the support of the CEO, resulting in a high response rate within a short time-frame.

To ensure data accuracy and efficiency, the survey was designed to collect precise numerical information. All questions were mandatory. The survey commenced with a general introduction before delving into specific sections covering *overall experience*, *utilisation*, *productivity*, *performance*, *features*, *challenges*, and *training*. Respondents who had not used Pixie were directed to the training and comments sections, streamlining the process and

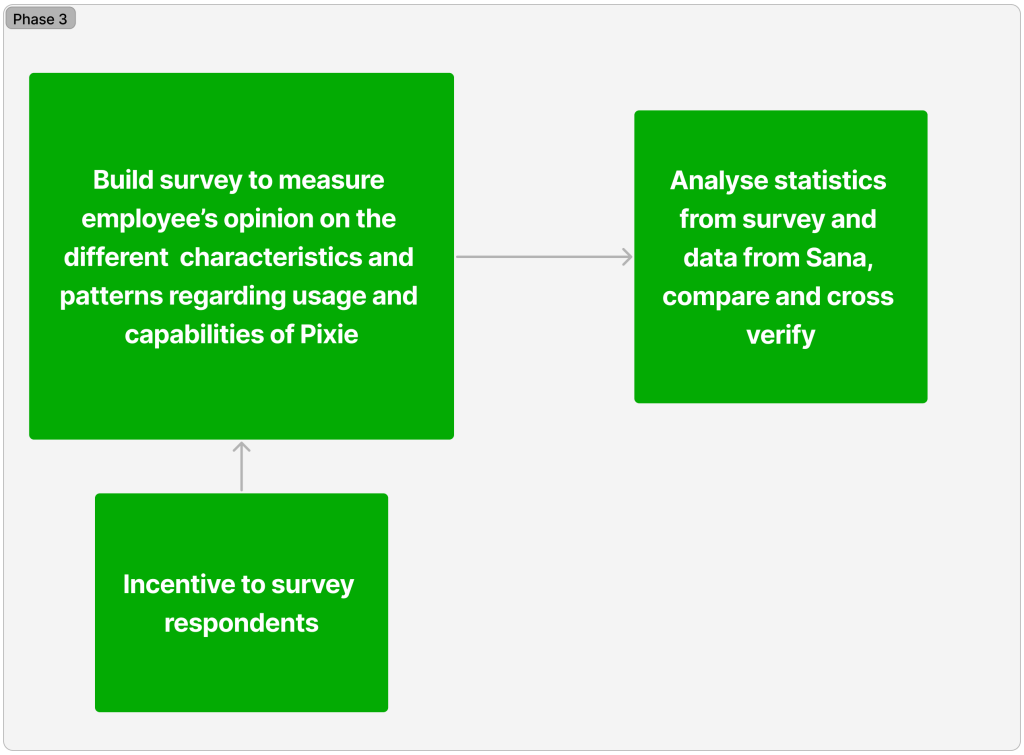


Figure 3.4.1: Phase 3 - Data collection through survey

preventing survey fatigue. This branching logic was intended to optimise the survey experience, which would lead to higher completion rates and responses.

Overall Experience

The Net Promoter Score Net Promoter Score (NPS) is a valuable metric traditionally used to gauge customer satisfaction and loyalty on a scale of 1-10 (Reichheld, 2003). While it is widely utilised in customer-facing contexts, this study adapts the NPS for internal use within the company to measure employee satisfaction with Pixie, an AI tool used by Piab employees. The NPS is calculated based on responses to a single survey question, where respondents are categorised as Promoters (scores of 9-10), Passives (scores of 7-8), and Detractors (scores of 0-6). Promoters are loyal enthusiasts who are likely to continue using the tool and recommend it to others, thereby fostering internal growth and adoption. Passives, though satisfied, remain unenthusiastic and vulnerable to alternative solutions. Detractors are dissatisfied users who may negatively impact the tool's perception through unfavourable feedback. The NPS is derived by subtracting the percentage of Detractors from the percentage of Promoters, resulting in a score that ranges from -100 to +100. Higher scores indicate stronger loyalty and satisfaction. By employing the NPS internally, the aim is to gain insights into the overall employee experience

with Pixie, identify areas for improvement, and enhance its utility within the organisation.

Utilisation

Utilisation aims to capture information about the frequency of usage (daily, weekly, monthly) as well as the common uses and purposes of the chatbot. By examining the frequency of usage, this was used to compare it against various metrics to uncover insights about usage patterns, consistency of engagements, and trends over time. Understanding the purpose of usage helped to identify what has been working well for employees, highlighting the most commonly used features and the least utilised ones, thereby providing a distribution that shows the relative importance of specific features. Additionally, cross-tabulation was performed to indicate which function groups find specific features more useful than others, offering quantitative insights that complement the qualitative data. Should historical data have been available, it would be possible to observe trends over time.

Følstad, Skjuve, and Brandtzaeg (2019) discuss that user-driven chatbots, such as personal assistants, frequently demonstrate high engagement consistency due to their integration into daily routines and personalised interactions. In contrast, chatbots designed for short-term, specific tasks, such as customer support or information retrieval, may exhibit more variable usage patterns contingent on immediate user needs. By systematically tracking both the frequency and context of interactions, as well as analysing qualitative feedback, organisations can garner comprehensive insights into chatbot utilisation, identify key features, and discern usage trends over time. This approach supports our methodology for survey data analysis, ensuring a thorough understanding of how Pixie is used within the company.

Productivity

In a study conducted by Brandtzaeg and Følstad (2017), it was found that the primary motivation for using chatbots among the study group was productivity. Users appreciated the ease, speed, and convenience of obtaining assistance or information quickly. Many users preferred chatbots over human assistants due to the immediate and non-judgemental responses. For chatbots to be effective, they must enable users to complete tasks or achieve specific goals efficiently and effectively, ensuring ease, speed, and convenience. Additionally, they must meet crucial productivity needs, such as providing immediate assistance or information Brandtzaeg and Følstad (2017).

The productivity section of the survey aims to demonstrate the time saved and quantify the

amount of time saved by Pixie users. Key statistics that can be calculated include the *average* time, *median* time, and *mode* time saved. Analysing the distribution of time saved allows for the identification of clusters where most users fall, as well as any outliers. Categorising this data created a view to see the proportion of users experiencing varying levels of productivity gains. Measuring the time saved by users highlighted the tangible benefits of using Pixie, thus providing a strong case for its value proposition. These quantitative results were further analysed in conjunction with qualitative insights to understand how Pixie impacts users' work, which specific features are particularly beneficial, and areas that may need improvement.

Performance

Collecting these performance metrics was essential to understand how they influenced user engagement with the chatbot. This data helped identify strengths and weaknesses, pinpoint areas for improvement, and direct resources to where specific enhancements were needed. Areas with lower scores required more attention and improvement, while areas with high scores could be optimised further or used as models for other features. These ratings were also aligned with the features that contributed most to productivity, thereby guiding the development of new features or enhancements of existing ones.

Under this category, many of the questions related to the quality of the chatbot. The quality aspect was broken down into specific questions that candidates could rate. Quality encompassed metrics such as *validity*, *accuracy*, *reliability*, and *adaptability*. Validity measured how relevant and clear the responses were based on the user's prompt and needs, which was inherently subjective. Accuracy was quantified by the percentage of responses that were correct or appropriate given the user's intent. High accuracy in a chatbot ensured that users received relevant and correct information, enhancing user satisfaction and efficiency in handling inquiries or problems. This metric was also subjective as users did not typically provide feedback on the accuracy of the data and its usage. Reliability referred to the consistency of responses to repeated or similar prompts. Asking about adaptability provided subjective information on how well Pixie adapted to users' specific roles and needs.

Features

Questions in this section aimed to uncover the effectiveness and reception of various features. These inquiries were designed to gather positive feedback and experiences, which could then serve as credible sources for internal marketing of the tool. Additionally, the information obtained was used to highlight certain features and identify potential areas for

future development and growth. Management could use these insights to focus on enhancing features and improving performance aspects such as quality and speed. Furthermore, this section sought to identify features that users desired, which could potentially increase their productivity and simplify tasks.

Research indicates that users value chatbots for their ability to provide immediate assistance, automate routine tasks, and offer personalised interactions (Brandtzaeg and Følstad, 2017). Features such as NLP for understanding user queries, integration with other tools and platforms, and real-time data access are critical for improving user satisfaction and productivity (Abu Shawar and Atwell, 2007).

Challenges

Challenges are crucial for development and improvements, as they help in understanding the needs and pains of users, enabling actionable steps to address concerns. Identifying challenges can uncover a wealth of insights that are essential for targeted improvements and strategic planning. It is important to determine whether all function groups face the same challenges or if they are specific to roles within function groups. This could indicate that role-specific customisation's and adaptations of Pixie are required in certain areas. The challenges highlighted by respondents will provide a valuable comparison with the performance of the tool. Some challenges may relate to speed, suggesting that technical enhancements are necessary, as these issues can negatively impact user satisfaction and productivity. If there are challenges related to reliability, users may trust the tool less, leading to decreased usage. Identifying and matching challenges with accuracy can reveal that the quality of the information is not up to standard, which can then be compared with the qualitative results.

Training

Since training and development are essential parts of learning and understanding how to utilise a new system, whether physical or digital, these questions were absolutely necessary in the survey. This approach ensured that each respondent only saw the questions pertinent to their own experience, customised to flow smoothly, thereby increasing the likelihood of obtaining accurate responses and a higher response rate. The results helped identify if there was a gap in internal communication and training reach. Additionally, it gauged interest in potential training opportunities and the type of training experiences respondents preferred. These insights will help the business understand the necessary investment required to implement training, determine if it should be better communicated, or both.

3.5 Phase 4 - Recommendations and Refinement

In this phase, as shown in Figure 3.5.1, observations and results from the preceding phase are used to construct recommendations. These recommendations encompass several key components: proposed improvements in focus areas, a consolidated analysis of the interviews, surveys, and other collected data, potential next steps, identification of possible future risks along with mitigation strategies, and any necessary integration and transition steps. This was an iterative process, and not only was there a substantial amount of data to analyse, but it also required business and people knowledge to extract insights that are valuable to the organisation. For additional reference, the entire flow chart of the research process can be found in Appendix B.

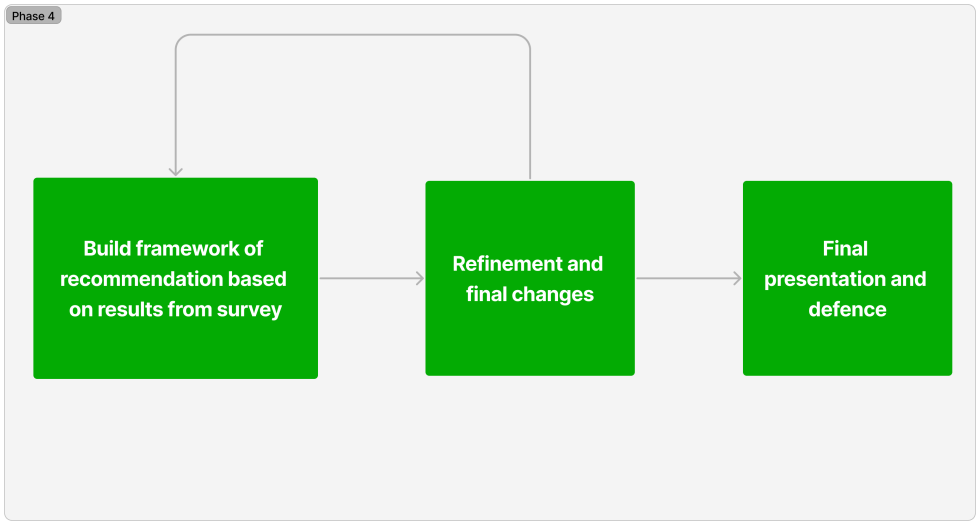


Figure 3.5.1: Phase 4 - Recommendations and refinement

3.6 Methodological assessment

The validity and reliability of the data used in the Power BI dashboards were critical, as these dashboards were the primary source of information and metrics from Sana. It was necessary to check the data weekly to ensure there were no discrepancies or missing data. On a few occasions, it was noticed that some data was missing or incorrect; these issues were resolved promptly to avoid delays in data processing and analysis. The validity of the data depended on the pre- and post-processing in Power BI. It was trusted that the files provided had the correct equations and formulas to sort the data. A significant amount of learning was required to understand the system, and there were times when a few formulas needed adjustments.

Ensuring the accuracy of these formulas was vital for maintaining data integrity.

Given the scope of the research, it was impractical to validate all the data comprehensively. Therefore, a targeted approach was employed to ensure that only the necessary data relevant to answering the research questions was validated. This selective validation process focused on key metrics and data points that directly contributed to the research objectives. Data that was not relevant to the research questions was deliberately excluded from the validation process to streamline efforts and maintain focus on the most critical information.

By using multiple sources of information, triangulation was employed to cross-verify the findings and this enhanced the the validity of the research. This approach also increased the confidence in the results presented and reduced the risk of bias. This was done in Power BI, where the data from interviews, the survey, employee details and chatbot data were combined into tables, and charts and graphs were created to provide visual representations of specific departments and groups, based on the criteria and requirements that were put forward to investigate in the research.

Data management practices combined with consistent data collection from the weekly reports ensured reliability in the research. Throughout the research, the data coming in was thoroughly checked and compared to previous data sets, ensuring that there were no discrepancies. If any issues or inconsistent data was found, it was reported and fixed, ensuring that the procedure and process for the next iteration would be smooth. It was only necessary to have more that one person view and interpret the data and visuals to get a different perspective. This practice ensured that the interpretations were consistent and reliable. Furthermore, the study adhered to ethical standards by ensuring confidentiality and informed consent from all participants. This ethical consideration contributed to the reliability of the responses, as participants were more likely to provide honest and accurate information.

In summary, the methods employed to ensure validity and reliability included regular data checks, targeted validation of relevant data, triangulation, consistent data collection procedures, and adherence to ethical standards. These measures collectively ensured that the findings of the study were robust and trustworthy.

3.7 Research Bias

At the start and during the course of the study, steps to avoid any form of bias were always considered and taken at necessary stages. One of the main biases that were predicted as a

potential occurrence was selection and sampling bias. Šimundić (2013) says that selection bias occurs when study participants are not representative of the broader population that the research is intended to examine. The results may be distorted by this lack of representation, rendering the conclusions invalid for a larger population. She also says that, closely connected to this bias is sampling bias. She defines mistakes that could arise during the sample selection procedure and compromise the sample's representation. Another bias that could arise is in the data interpretation. She explores how scientists could keep interpreting data to validate their theories even in the absence of statistical significance. Confirmation bias can be detected in this way, whereby differences and relationships that are observed are discussed as though they are important when they are not.

- **Diverse Participant Recruitment**: We ensured a diverse and representative sample by recruiting participants from various departments and roles within Piab. This was done to minimise selection bias and ensure that the study results are generalised to the entire user population.
- **Random Sampling**: Participants were randomly selected to participate in the interviews and surveys, reducing the risk of sampling bias and ensuring that the sample accurately represents the broader population.
- **Blinded Data Analysis**: Data interpretation was carried out with blinded analysis where possible, to prevent confirmation bias. Analysts were not aware of the specific hypotheses being tested, which helped in maintaining objectivity in data interpretation.
- **Standardised Procedures**: A standardised interview and survey protocol was followed to ensure consistency in data collection across all participants. This included using the same set of questions and procedures for every interview and survey.
- **Pilot Testing**: The interview and survey instruments were pilot-tested to identify and rectify any potential biases or ambiguities in the questions.
- **Cross-Verification**: We used triangulation by combining qualitative and quantitative methods to cross-verify the findings. This approach helped in corroborating the results and reducing the impact of any single method's bias.
- **Feedback and Iteration**: Feedback was sought from participants and external reviewers to identify any overlooked biases and refine the research methods accordingly.

Interviewees	**Function**	**Messages Sent**
Interviewee A	Business Development	68
Interviewee B	Inside Sales	347
Interviewee C	Inside Sales	3
Interviewee D	Management	1121
Interviewee E	Management	210
Interviewee F	Management	134
Interviewee G	Management	119
Interviewee H	Management	3
Interviewee I	Marketing and Communications	1861
Interviewee J	Marketing and Communications	1163
Interviewee K	Operations	426
Interviewee L	Operations	121
Interviewee M	Operations	3
Interviewee N	Operations	1
Interviewee O	Outdoor Sales	545
Interviewee P	Outdoor Sales	348
Interviewee Q	Outdoor Sales	207
Interviewee R	Outdoor Sales	18
Interviewee S	Outdoor Sales	1
Interviewee T	Outdoor Sales	0
Interviewee U	R&D	992
Interviewee V	R&D	440
Interviewee W	R&D	3
Interviewee X	R&D	2

Table 3.3.1: Extract from the list of interview candidates dated 11 March 2024

Groups in Piab	**Subgroups**	**Count**
Functions	Business Development	1
	Engineering	0
	Inside Sales	2
	Management	5
	Marketing and Communications	2
	Operations	4
	Outdoor Sales	6
	R&D	4
Divisions	VA	15
	VC	0
	LA	0
	China	3
	Shared Resources	1
	GM	5

Table 3.3.2: Interviewee numbers per Function and Divisions

Empirical Findings

This chapter presents the practical results of the study. It begins by explaining how data was gathered, processed, and analysed to be visually represented using Power BI. The data collection methods included several interviews conducted with a wide range of employees from different divisions and countries, as well as a questionnaire. Any changes made during the data collection process and the reasons for these changes are also discussed. The chapter then outlines the main topics identified from analysing the collected data and concludes by presenting the actual data itself.

4.1 Power BI Dashboards

In order to understand and interpret the usage (measured as the number of chat messages sent), usage patterns and adoption rates (number of users) of Pixie, it was essential to systematically collect and process relevant data. The data collection and processing period for the report was restricted from August 2023 up until May 2024. For the interview selection process as described in Phase 1, the data used was from August 2023 up until 10th March 2024. Further analysis and results came from the data up until May 2024, thus enriching the overall insights and results. The number of chat messages increased by 97% from 10th March 2024 onward. The usage and the adoption can be seen in Figure 4.1.1. The data is split by year and month for a better visualisation. A total of 87866 chat messages were sent through to Pixie.

Taking a closer look at specific months of increased usage, it is also worth noting that some insights were derived from informal insights gathered during weekly meetings and discussions with management. For example, the increase in usage In December was from talk that the CEO gave about using Pixie. It was also discovered that the reason for an increase in usage in March was due to the China team conducting internal training on Pixie. These informal insights were valuable in providing context and understanding behind certain data trends that might not have been immediately evident from the formal data collection methods. The number of users (adoption) accounts for 84% of the company and their target was 80%, however, it must be noted that the staff count was 1230, less the people in warehouse and manufacturing who aren't using Pixie.

The visual representation of users and non-users per function at Piab provides a clear and detailed breakdown of employee engagement with the Pixie AI chatbot as shown in Figure

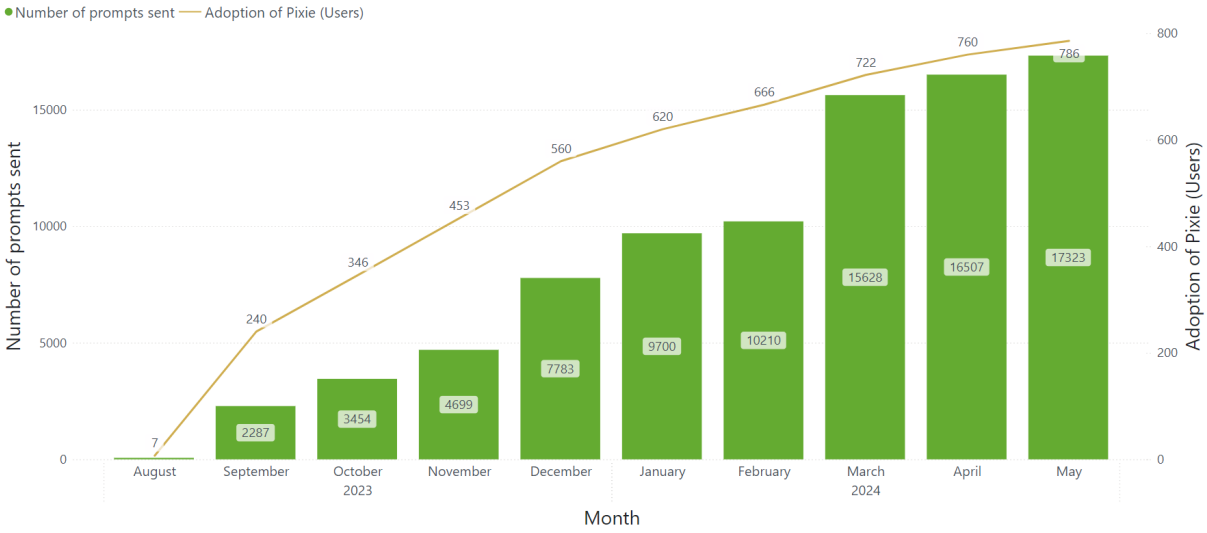


Figure 4.1.1: Monthly trend of chat messages sent and adoption

4.1.2. Each function corresponds to a specific department within Piab, and the chart categorises employees into two groups: those who actively use Pixie and those who do not. This segmentation is crucial for understanding the adoption rate of the tool within various parts of the organisation. This data is essential for driving strategic decisions to enhance productivity and ensure the effective integration of AI tools across the company. It must be noted that the decision to target specific groups was made early in the project, and not all functions are used in the analysis and discussion.

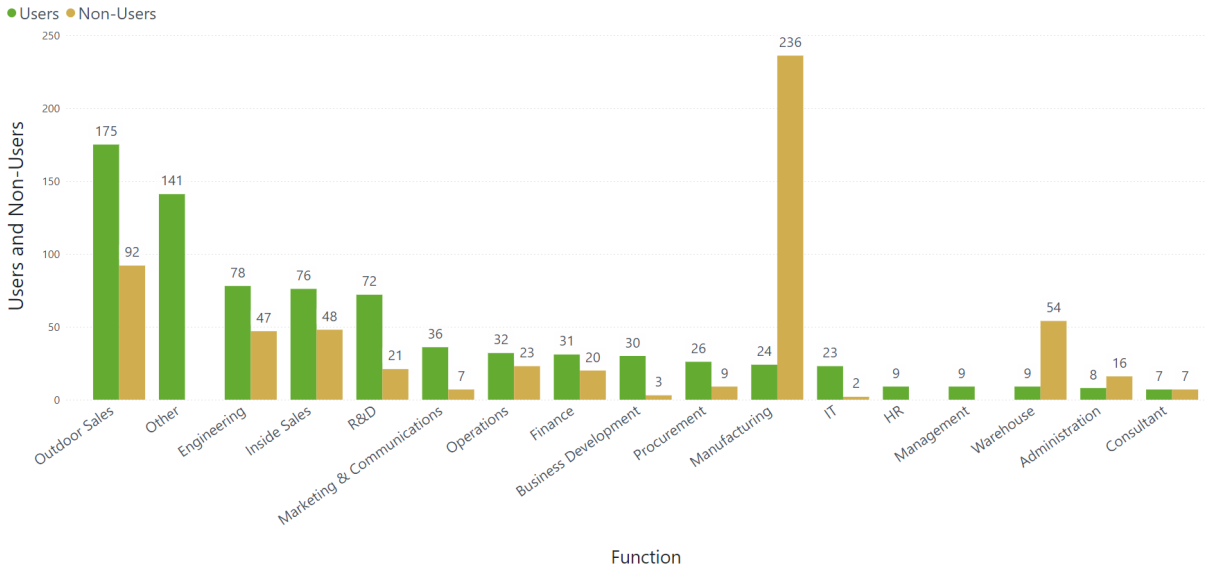


Figure 4.1.2: Users per function group

Additional information regarding the statistics concerning the messages sent in various countries, and the number of messages sent versus the number of users can be found in

Appendix C. This data is provided for informational purposes to showcase that other angles and approaches can be used to visualise and further analyse the data, depending on the needs and goals of the company.

In this research project, the above-mentioned charts were pivotal as they contained the relevant data necessary for triangulation in the analysis. Triangulation, as discussed earlier, involves using multiple data sources and methods to cross-verify findings, thereby increasing the accuracy and reliability of the results. By incorporating statistics on message distribution across different countries and user engagement levels, the study ensures a comprehensive understanding of usage patterns and potential areas for improvement.

Furthermore, this additional data can be leveraged by the company for various strategic purposes, such as identifying regional trends, understanding user behaviour, and tailoring training and support initiatives to specific user groups. By providing a multifaceted view of the data, the company can make more informed decisions and develop targeted strategies to enhance the effectiveness and adoption of Pixie.

Overall, while the primary focus of this research was on the key metrics and data points relevant to the research questions, the supplementary data in the appendix offers valuable insights that can be explored further to support ongoing and future initiatives within the organisation.

4.2 Interviews

The interviews were conducted with a diverse group of individuals, ranging from new users to frequent users, across different divisions within the organisation, various functional groups, job titles, and countries. The interview process was meticulously organised and structured, adhering to a prearranged interview guide. Interviews were conducted in person for those located in Stockholm, Sweden, and virtually for those located offshore, to gather insights and information on their experience using the chatbot. The interviews also aimed to collect information about potential features that could be added to Pixie to make it more appealing, accessible, accurate, and reliable for employees.

4.2.1 General findings

The candidates were selected based on the organised data structure and the needs of Piab. Candidates were selected from eight out of the sixteen function groups within Piab from the dataset, although not all selected candidates were interviewed. Despite numerous emails

sent and assistance from management, it proved challenging to schedule all the meetings. Contributing factors included time constraints, some individuals being unavailable during the planned interview period, others being off sick, or some not wanting to participate in the study. This resulted in a total of 24 interviews being conducted.

”Pixie was introduced as an assistant that can summarise my meetings & draft essays or emails.”

The interviews mentioned in section 3.3.3 were analysed through coding using the Attride Stirling Thematic Network tool (Attride-Stirling, 2001) to structure the findings from the interviews and compare them. All interviews were transcribed and then they were individually coded. The basic themes were important and relevant keywords or phrases that interviewees mentioned. These keywords and phrases were extracted from the transcription and from the recorded interviews. Initially, there was a standard and simplistic set of organising themes that were linked to the basic themes. This initial list was created as a starting point and underwent several iterations as we conducted more interviews and gathered additional information. The organising themes also had to be relevant to the research questions to ensure that when the three global themes were formulated, they would be coherent and assist in answering the research questions. The basic themes were then categorised and sorted into these evolving organising themes. The Global and organising themes can be seen in Table 4.2.1, and all the results that include the basic themes can be found in Appendix D.

”It is like an assistant.”

”Pixie is like your personal help or assistant that makes your life easier and you can use it in a smart way to be more efficient”

The results from the interviews were organised methodically to ensure an easy flow for the interviewees. The global themes — User Interaction and Experience with Automated Tools; Training, Challenges, and Impact on Job role; Future Enhancements and Digital Transformation — were instrumental in organising the insights gathered from the interviews. These themes helped to categorise the major common points as well as unique comments specific to departments. By grouping the recurrent and unique points under these global themes, we were able to formulate the takeaways from the interviews, providing a structured output for the qualitative section of the study. This thematic organisation ensured that the results were comprehensible and directly aligned with the research questions, thereby offering a coherent

Global Themes	Organising Themes
User Interaction and Experience with Automated Tools	Document summarising and analysis Translation and Language support Data retrieval and Search function Meeting summaries and note taking Integration with software application tools used in the organisation Content drafting and structuring assistance
Training, Challenges, and Impact on Job role	Training and user support Technical limitations and accuracy Adoption and cultural resistance Integration, accessibility and presence Language and translation limitations Data and content relevance Reliability, Accuracy and Trust
Future Enhancements and Digital Transformation	User interface Improvements Advanced Analytical features Mobile app development Customisation, Personalising and Focus options Image and visual data processing Voice interaction and search features Security & Data privacy enhancements Integration with external platforms and tools Language and translation improvements

Table 4.2.1: Global Themes and Organising Themes

analysis of the interview data.

4.2.2 User Interaction and Experience with Automated Tools

The usage patterns of individuals can be split into three groups - Daily, Weekly, and Monthly -as per the qualitative results from the interviews. The count of users per group is depicted in Table 4.2.2

Usage	Count
Daily	12
Weekly	7
Monthly	6

Table 4.2.2: Usage patterns of people

Document summarising and analysis

This feature empowers users to upload their content and receive concise summaries. It offers

quick insights into the material, aiding in efficient document comprehension and understanding. The feature has proven highly popular amongst employees, with many reporting substantial time savings in document analysis. Users can upload content as text or documents (PDF, Excel, Word, etc.), which Pixie, can process to generate summaries or answer specific queries. It's a good tool that can be used to correct grammatical errors, spelling, and create either professional or simple sentences when communicating Piab info to others, or for learning.

Translation and Language support

The translation tool is widely utilised by employees engaging with global customers and suppliers, and internally. It is particularly valuable for colleagues in overseas offices who require accurate translations of documents, emails, and product information into local languages. This facilitates effective communication and collaboration with external partners and customers. The tool is instrumental in handling the significant volume of messages requiring translation. It creates a foundation for employees to better understand the products, services and documents of Piab.

Data retrieval and Search function

Employees frequently employ the search tool to locate information about Piab products and their technical specifications swiftly. It is especially beneficial for roles that lack technical expertise but occasionally need to reference or communicate product-related details. While useful for technical staff, the tool's accuracy and consistency in delivering results can be improved.

Meeting summaries and note taking

Employees value the meeting transcription, note-taking, and the summary feature as a supplementary tool for tracking meeting updates and decisions. It assists in managing information from attended and missed meetings.

”Pixie was introduced as an assistant that can summarise my meetings & draft essays or emails.”

Integration with software application tools used in the organisation

This feature enhances the efficiency of document analysis and data retrieval. However, its capabilities are often overlooked, and there is potential for wider utilisation and development. Employees familiar with the tool suggest areas for improvement and expansion. At the time of the study, it was integrated with microsoft teams for recording and summarising meetings and partially integrated with salesforce. But from the interviews, it was understood that the although many users who tried to use it found it useful but not consistent with its performance,

a few users weren't sure how to use the integration feature.

Content drafting and structuring assistance

Many employees have found this feature helpful in improving the efficiency of content creation for presentations, and internal and external marketing content. By providing user-specific guidance, the tool assists in structuring and drafting various content types, from emails to product descriptions, presentation agendas, and content ideas for marketing.

4.2.3 Training, Challenges, and Impact on Job role

Training and user support

Many interviewees found this topic beneficial and saw potential for improvement. Piab had already uploaded training videos on Pixie's features and usage on the intranet at the time of the interviews. While most knew about these resources, some did not, and perhaps the communication about the material could have been improved. Many preferred shorter training formats or in-app ”training snippets” over videos or documents.

Employees in offices outside Sweden highlighted language as a barrier to accessing training materials, as these were primarily in English. Additionally, some interviewees struggled to keep up with Pixie's updates and new features.

To enhance training, communication, and user support, interviewees suggested:

- Providing training videos and documents in all relevant corporate languages.
- Integrating training snippets and helpful prompts into the user interface.
- Organising internal knowledge-sharing sessions on Pixie best practices and use cases.
- Regularly communicating about new Pixie updates and features.
- Offering concise training options for new users.

Technical limitations and accuracy

Interviews revealed that employees experienced inconsistencies in Pixie's responses. These included inaccurate technical product information, failure to recognise Piab terminology, delayed or absent replies, broken source links, coding errors, and overly broad or irrelevant information. Many employees also reported slow response times and Pixie's occasional inability to remember recent prompts and answers. Additionally, Pixie's inability to process images hindered users from summarising content with graphical or pictorial data.

”It sometimes provides a rock solid answer which is wrong”

Adoption and cultural resistance

It was understood from the interviews that many of the users who were completely new to using AI and/or did not understand how the tool works felt an internal resistance towards using the tool. Although most felt that they should be more accepting of the new technology, they reported to have felt hesitant towards delving into using an AI tool.

Integration, accessibility and presence

The integration feature is undervalued as some interviewees were either unaware of the feature or did not know how to use it. Some interviewees felt that the User Interface (UI) was difficult to get use to and found it difficult to find where certain features were. Coincidentally, a lot of the users also mentioned that they may be finding it difficult because of their age or because Pixie is new to them or they are new to using an AI chatbot.

The opinion about Pixie’s presence at different interfaces that employees use were varied. While some users liked that Pixie was readily available for use in their browser and would show up during online meetings by default, others did not prefer that and felt an ”opt in” feature should be there instead of an ”opt out”, especially with integration in online meetings. The latter group felt the user should be able choose whether Pixie enters the meeting rather than having it to leave the meeting.

Several employees also mentioned that the current software integration capabilities, although impressive, are not entirely relevant or useful for their roles. They felt that deeper integration into their role-specific tools would be preferable and would significantly enhance Pixie’s value to them. (E.g.: Integration with Microsoft Office tools, analysis from Salesforce data).

Language and translation limitations

Although the feature is widely used and appreciated by many employees it isn’t devoid of some issues and limitations. Some interviewees mentioned that translations results seem to reduce in effectiveness when the direction of translation is English to the local language. Translation of complex technical terms in both directions (English <-> Local Language) has also been found to be inaccurate by many.

It was also reported in many interviews that the language model that Pixie identifies & presents results in is:

- Sometimes too generic.

- Not identifying that the user is an employee of Piab.
- As though Pixie is external to Piab.
- Sometimes doesn't recognise Piab terminologies.
- Uses wrong terminologies despite correct prompts from the user.
- Doesn't seem to correct it's responses despite correction given by the user.

Data and content relevance

Interviews revealed that many employees encountered instances where Pixie provided outdated or irrelevant information in response to their queries. While the nature of the prompt might influence the relevance of the data, it raises questions about Pixie's ability to understand the context of Piab-related prompts. Technical and sales roles reported inaccuracies, insufficient detail, or low confidence in product information provided by Pixie, compared to their own knowledge and expertise. Some interviewees also described "hallucinations" in Pixie's responses, likely caused by inconsistencies within the vast datasets used for training the RAG models used for indexing multiple documents with conflicting information.

Reliability, Accuracy and Trust

There were also concerns regarding trust & confidence in Pixie and it's results raised by the interviewed employees. It was understood from the interviews that:

- There are underlying confidence issues with the product-related suggestions and results provided by Pixie, primarily due to the format of referencing or citing the source information. The references provided were either invalid links or entire documents. The problem with referencing an entire document is that, if it contains multiple pages, it becomes difficult for the user to identify the specific source of the information, making it challenging to cross-check and verify.
- There are a lot of questions, doubts, and trust issues surrounding the ownership of data collected and presented by Pixie. Through currently available training videos, it has been clarified that data collected by Pixie is owned by the Piab Group and secured by Sana AI. This still made some users feel uncomfortable speaking freely with Pixie present in meetings as they were unsure who would view the recorded data and how it would be used.
- There were also reliability issues when it came to Pixie's product-related results for Coval (an entity with different branding and product names, but also a competitor) (Refer to

Figure 1.1.1). The issues centred around the unavailability of Coval product data; users often received only Piab product suggestions. This forced users to consistently upload catalogues they had immediate access to, undermining the purpose of the product search feature. As a result, the tool's product suggestion and search function was unreliable for Coval users.

- Few users resorted to using their own RAG model to serve their requirements as they found it to be more accurate than depending on the model inbuilt in Pixie. This is probably due to the vast set of data that the inbuilt RAG model indexes making it difficult for users to prompt Pixie to find and focus on specific set of documents that are suited to their need.

4.2.4 Future Enhancements and Digital Transformation

User interface Improvements

It was understood that some improvements in the interface could be made regarding intuitiveness, integration with external tools, and the selection of documents from the Piab database through Pixie (file/file collection tagging), especially for new users. This would further encourage the adoption of Pixie within the company. Additionally, the following points and the survey results provide specific insights on how to improve the user interface.

Advanced Analytical features

This suggestion, closely related to improved integration, specifies the exact integration required. It was mentioned by interviewees whose roles involved working with large amounts of analytical data. They requested that Pixie be able to perform basic initial analysis based on user prompts.

Mobile app development

A small group of interviewees (3 out of 24) suggested that a mobile app version of Pixie would increase usage. They felt, Pixie being available as a mobile app would make them use more.

Customisation, Personalising and Focus options

Follow-up discussions with interviewees identified a tendency for Pixie to provide excessive explanations and data, hindering efficiency. To address this, interviewees suggested an option to personalise Pixie. They proposed allowing users to tailor Pixie's behaviour based on their role, focusing its output on specific documents and data relevant to their work. A 'role-based focus' feature would facilitate this personalisation.

Image and visual data processing

Through the interviews, it was understood that in order for extracting the maximum potential of the document summary feature, users hoped for Pixie to be able to read, analyse and summarise visual data. Along with this, users associated with content creation roles also requested for prompt based image generation capabilities to be provided by Pixie.

Voice interaction and search features

Voice interaction and searching feature was only mentioned by a few interviewees but was referred to be capable of framing prompts much easier for the few who mentioned it. It was not so popular and Piab had mentioned trying it in the early stages but it was still too slow at the time.

Security & Data privacy enhancements

Clearer communication of policies regarding usage of data collected by Pixie was something that was understood from the concerns and suggestions raised regarding data privacy, usage and ownership. Effort in this aspect would help employees feel more comfortable and secure using Pixie freely and to its fullest potential.

Integration with external platforms and tools

Along with appreciating the availability of integration with external tools (Calender, MS teams meeting etc), interviewees also hoped for Pixie to be able to integrate with their role specific tools (Eg: MS Office, Power BI, Customer Relationship Management (CRM), ERP, ERM, etc)

Language and translation improvements

The general consensus was positive about the translation feature's presence and performance. But a few issues that were raised revolved around the effectiveness of translated material reducing when translation from English to local language is done.

4.3 Survey

A company-wide survey was conducted to record the responses from employees. The survey was designed to test the observations and collect key insights derived from the qualitative study and to avoid sampling and selection bias. Through the quantitative study, 320 responses were received out of 1230 employees to whom the survey was sent out, from which, at the time of the survey, about 773 employees were active users.

The response rate achieved was 26%, which included not only FTE employees but also

individuals outside this category, such as consultants, interns, and part-time staff. This diverse range of respondents provided a broader perspective on the usage and effectiveness of the Pixie tool, enriching the dataset with varied insights from different employment statuses. When analysing the data from the survey and combining the internal company data to have better representation of the function group, an overview of the divisions and functions are shown in Table 4.3.1. It can be seen that there was low representation among most of the groups.

Piab Split	**Subgroups**	**Count (% in Piab)**
Functions	Administration	5 (20.83%)
	Business Development	12 (36.36%)
	Consultant	0 (0.00%)
	Engineering	25 (20.00%)
	Finance	13 5.49%)
	Human Resources (HR)	5 (55.56%)
	Inside Sales	42 (33.87%)
	IT	12 (48.00%)
	Management	3 (33.33%)
	Manufacturing	16 (6.15%)
	Marketing & Communications	17 (39.53%)
	Operations	15 (27.27%)
	Outdoor Sales	94 (35.21%)
	Procurement	9 (25.71%)
	R&D	37 (39.78%)
	Warehouse	1 (1.59%)
Divisions	VA	15
	VC	0
	LA	0
	China	4
	Shared Resources	2
	GM	5

Table 4.3.1: Results of Function and Divisions from survey

It must be noted that a small percentage of users and respondents fall under a special category and will be grouped as ”Other”. These users are either new to the company and have not been added to the Full Time Employment list, internship and master’s thesis students, people that have left during the research period and consultants that have a Piab account. These percentages will be mentioned in the results when necessary and applicable.

For the ratings, and even number system was used to eliminate neutral responses and it also simplifies the overall analysis making it easier to identify trends and patterns in the results. When respondents are forced to choose between positive and negative options, they are compelled to express a preference or opinion, and thus provides more actionable data for decision-making processes.

The results are presented below and has been divided into 7 different sections as mentioned in section 3.4.1. The sections created made it easier to categorise the data for interpretation and discussion with qualitative data.

4.3.1 Overall User experience

To understand the level of awareness of Pixie's existence in the company and its usage, questions asking if they know of Pixie and if they use it were posed. 95% of respondents are aware of Pixie from which 83% use it. The 2 leading causes for those who are aware of Pixie but don't use it is *no relevance of Pixie to their job function (39%)* and them *being unaware of Pixie's capabilities (30%)*. The remaining causes, 31%, were spread across *privacy and trust concerns*, *technical limitations* and other reasons mentioned by respondents such as not having the time to try it out.

The sections that follow contain the responses from the 251 people who use Pixie, except for the training section, which includes responses from everyone. This is due to the branching system that was used in the survey as mentioned in section 3.4.1. Respondents rated their overall experience with Pixie on a six-point scale, resulting in an average score of 4.63. This indicates a generally positive perception of the service, with 60% of respondents leaning towards higher satisfaction levels.

It is important to know what percentage of the users are promoters of the tool to understand whether it can be promoted by word of mouth within the company. It is equally important to know about those that don't see the value in the tool. To do this, the concept of NPS was used as discussed in section 3.4.1. It quantifies the likelihood of a customer (user in this research) to recommend the product/service they are using to a friend/colleague (Baehre et al., 2022). A question was posed in the survey, directed towards those who are "aware of Pixie and use it", asking them to rate the likeliness of them recommending it to a friend or colleague, between 1 and 10. Based on the NPS logic, only 41% of the group gave a promoter rating while the detractors were 22% of the sample size. After the survey was closed, the final NPS was 19.

4.3.2 Usage Frequency and Purpose

To analyse employee usage patterns, data was collected on the frequency and purpose of Pixie utilisation, mirroring the qualitative study. The findings indicate that 29% of users accessed Pixie daily, 47% weekly, and 22% monthly. Building on the common purposes identified in the qualitative study, respondents were presented with a list of options and the opportunity to suggest others. The distribution of responses is shown in Figure 4.3.1.

The data reveals that Pixie is primarily used for *General Search Function*, *Sentence, content creation & drafting* and *Translation & language support*. Additionally, it is valuable for summarising information and addressing product-related queries. Beyond the predefined options, employees reported using Pixie for software coding and debugging. They also provided more specific details about the types of searches conducted within the general search function category.

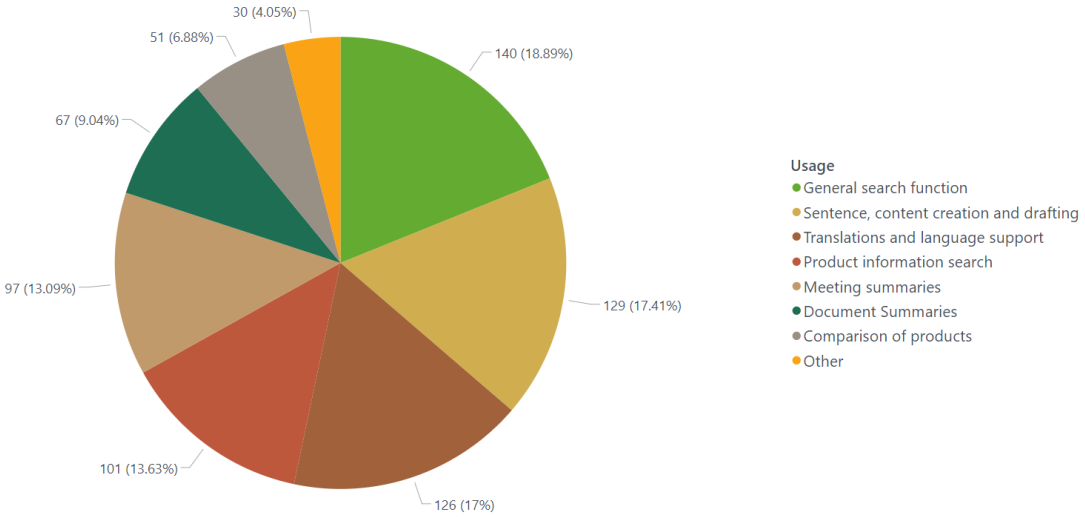


Figure 4.3.1: What is Pixie used for

4.3.3 Productivity

One of the first intentions of Piab with Pixie and introducing an AI chatbot tool for its employees was to help increase productivity and save time spent on repetitive, mundane but necessary tasks. To assess whether Pixie achieved this aim, we explored the relationship between Pixie usage and time saved. This involved asking questions about time saved when using Pixie and correlating these responses with training received.

77% of respondents who have used Pixie reported time savings over the past month. Conversely, the majority of those who did not report time savings also indicated a lack of training and

understanding. Furthermore, 69% of this group provided a low NPS, suggesting a correlation between training and perceived value. This implies that training plays a crucial role in enabling Pixie to enhance productivity, which in turn drives adoption.

To quantify the time saved by using Pixie, respondents were asked to estimate their time savings using a range of options. The distribution of these responses is shown in Figure 4.3.2. By taking the highest value from each range and calculating the total, it was estimated that Piab and its employees saved an average of 221 hours per week. This clearly demonstrates the effectiveness of providing Pixie to enhance employee productivity and suggests that this initiative should be expanded. When aggregated across the entire company, the average time saved for all Pixie users rises to an impressive 692 hours per week.

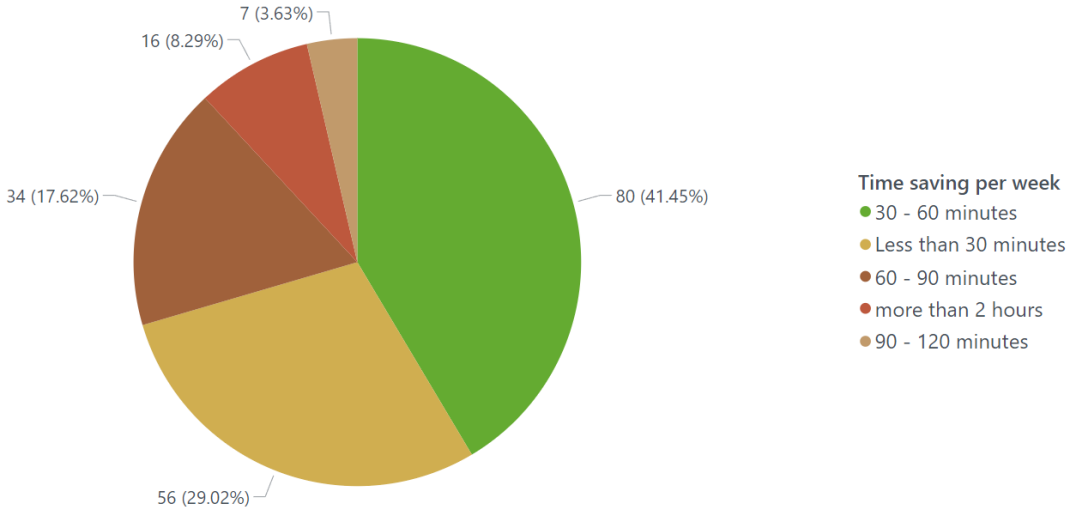


Figure 4.3.2: Survey results for amount of time saved using Pixie

4.3.4 Feature requests

Feature requests can be quite challenging to develop and adopt to a new technology that has been introduced. It needs to be features that bring value to the business through the quickest and most efficient methods. Here, 4 choices were given with and option to specify a personal request that was not in the list. The respondents were asked to choose 2 features they would like to see implemented in the future, and the results were surprisingly evenly distributed, as shown in Figure 4.3.3. This result further strengthens the output from the qualitative study that indicated popular feature requests.

Many users expressed interest in Image reading/Image creation, which suggests a need for capabilities that can analyse and generate visual content. Live translations in Teams meetings

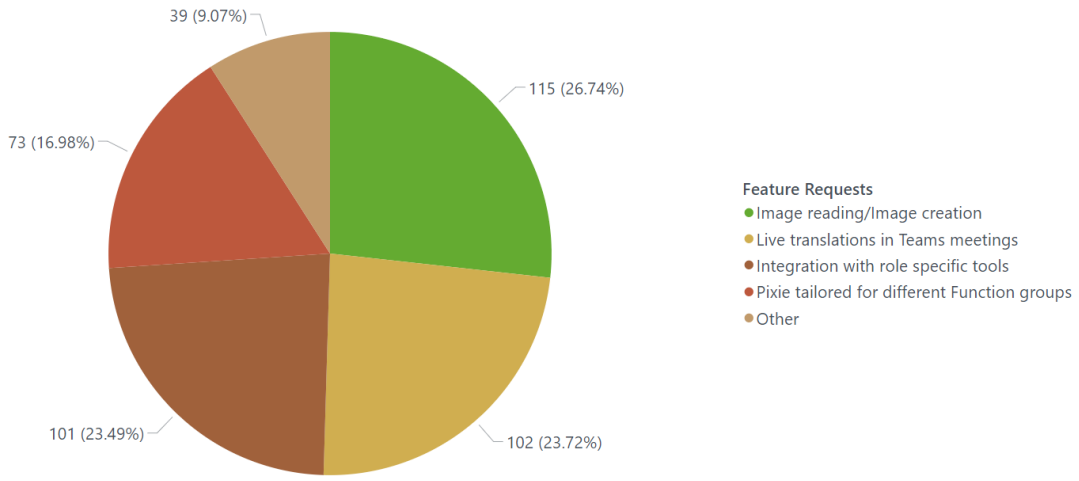


Figure 4.3.3: Survey results for most requested features

emerged as another prominent request, highlighting the importance of real-time language support in collaborative environments.

Additionally, there was significant interest in tailoring Pixie for different function groups, indicating that users value customisation based on specific departmental needs. Integration with role-specific tools was also a common request, emphasising the desire for seamless integration with existing software and tools used in various roles. Beyond the provided options, some users suggested personal requests, such as competitive information and comparisons, accuracy of data on Piab products, and enhanced product search capabilities.

There were also requests to integrate with other software such as CAD tools, ERP system, Sales Force and CRM. There were also a number of mentions about products which indicate a need on improving the accuracy, searchability, and comparative information of Piab products to better serve customer needs and enhance competitive positioning.

4.3.5 Performance

In order to understand how the users feel about the performance of the current features provided, a set of questions were asked in the survey to .

- **Usefulness of features**

  This question asked the respondents to choose features they find most useful. Shown in the figure 4.3.4 are the results. It can be seen that there is an almost even distribution between Summarising, Translating, Data retrieval and Content drafting & structuring features. It is interesting to note that, through the survey results, it appears that software

integration feature is currently not found useful by many. Through the qualitative study, it was understood that many found it difficult to understand how to use and experienced difficulties using it. The survey results reflect this very observation. Along with the previous question, the respondents were also asked to rate the usefulness of the features they selected. Shown in the figure 4.3.5 are the results from the responses.

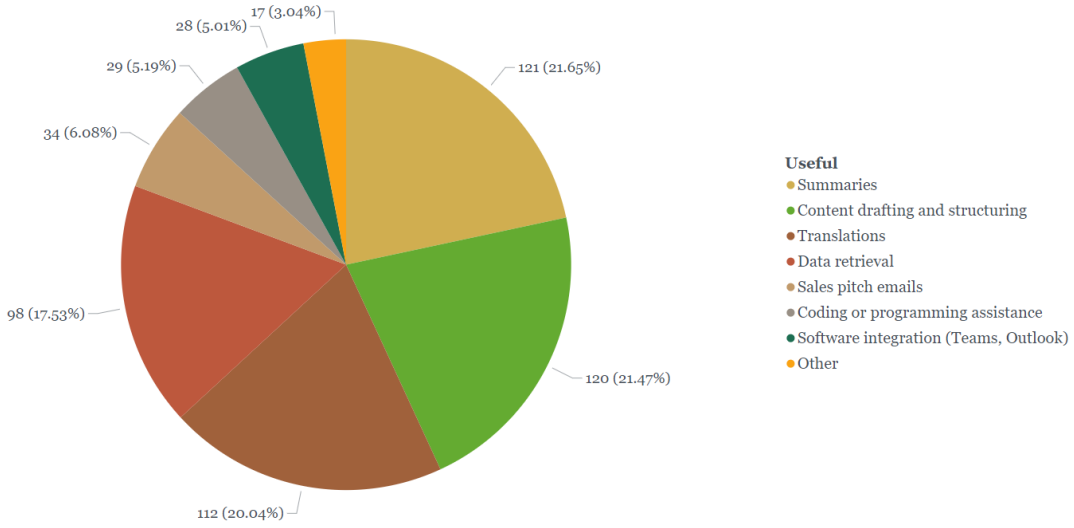


Figure 4.3.4: Survey results for most useful features

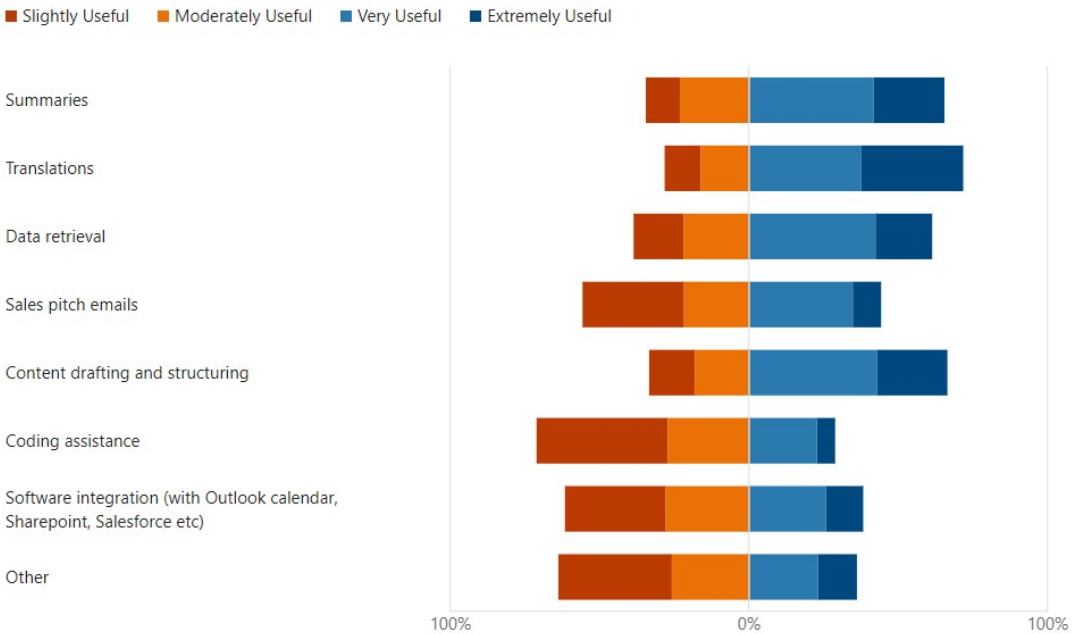


Figure 4.3.5: Survey results for rated useful features

- **Rating of Performance metrics**

  In order to have an understanding of how users feel about basic performance metrics associated with Pixie's responses, the respondents were asked to rate Pixie against those

parameters based on their experience. Shown in the table 4.3.2 are the results from the responses.

Performance metric	Average Rating
Validity of responses	4.58
Accuracy of responses	4.42
Reliability of responses	4.45
Adaptability of Pixie's responses to user's role and need	4.29
Level of trust in Pixie's responses	4.21
Overall confidence in Pixie's responses	4.31

Table 4.3.2: Rating of performance metrics

4.3.6 Challenges

Through the qualitative study conducted and the coding of interviews, frequently occurring challenges were narrowed down. To quantify the frequency of occurrence of these challenges and further strengthen the output from the qualitative study, survey respondents were asked to choose the challenges they have faced from a set list which were derived from the qualitative study. Shown in figure 4.3.6 is the distribution of the challenges faced by all the respondents.

Three challenges are observed to have a high count in comparison with others

1. Wong/Inaccurate answers to questions
2. Pixie giving too generic results to questions
3. Pixie entering meetings where I don't require it

There are also concerns about Pixie's translation capabilities, with reports of incorrect or inaccurate translations. The current integration of Pixie with external tools also appears to be a significant challenge, as users find these integration's not useful. Furthermore, Pixie's responses are often considered too generic, leading to low confidence in its results. Users also expressed discomfort with the current interface design and found the interface difficult to use.

In addition to these predefined challenges, several *other* comments were highlighted. These include requests for more visuals, pictures, videos, links, integrative maps, and directions, indicating a need for richer content and multimedia support. Some users mentioned issues with uploading files, particularly large ones, and errors in calculations with unit mistakes. There were also comments about the need for better meeting summaries and concerns about Pixie

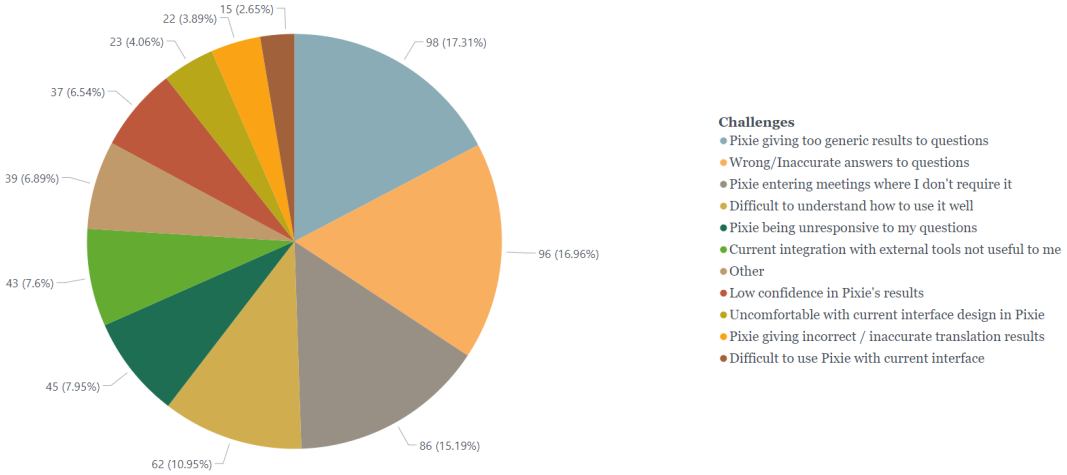


Figure 4.3.6: Survey results for challenges faced by users

starting or entering meetings incorrectly, even after they have concluded. A few users expressed that they have not yet had the time to explore all functionalities of Pixie or found it difficult to find real-time data on the internet and internal information.

These observations collectively provide a comprehensive understanding of the various user challenges and areas where Pixie can be further developed to meet user expectations more effectively.

4.3.7 Training and its Effectiveness

To understand the influence that training would have over use and adoption, some questions related to training, effectiveness and related preferences were asked. Through the survey it was understood that training only partially drives usage of the tool. 63% of the respondents replied ”No” when asked if they have received any form of training for using Pixie. Of these respondents, 100% of them use Pixie.

”We need to learn how to use Pixie and make the correct prompts!”

Respondents who replied ”Yes” to receiving training gave an average rating of 4.41 / 6 for the effectiveness of the training with 48% giving rating between 5-6. The small percentage of people who rated the effectiveness low, gave ”passive” rating for the NPS question. This indicates towards a possible correlation between training effectiveness and adoption, wherein, improving the effectiveness of the training could increase adoption of the tool.

57% of respondents are unaware of the availability of training videos on how to use Pixie, of which a majority of them (78%) interested in additional training material and sessions. The

remaining respondents, who are aware of the training videos, majority of them (64%) don't use the videos. The rest of the respondents, who have used the videos, have given an average rating of 4.5 / 6 for the effectiveness of the training/tutorial videos. This indicates that tutorial videos, although effective, has not reached the necessary audience to create an impact. 70% of the respondents expressed interest in additional training sessions or resources on how to to use Pixie better. As shown in 4.3.7, popular suggestions from the qualitative study was given as choices. The preference distribution shows Webinars & Live demonstrations and Online Tutorials & E-Learning as the popular choices.

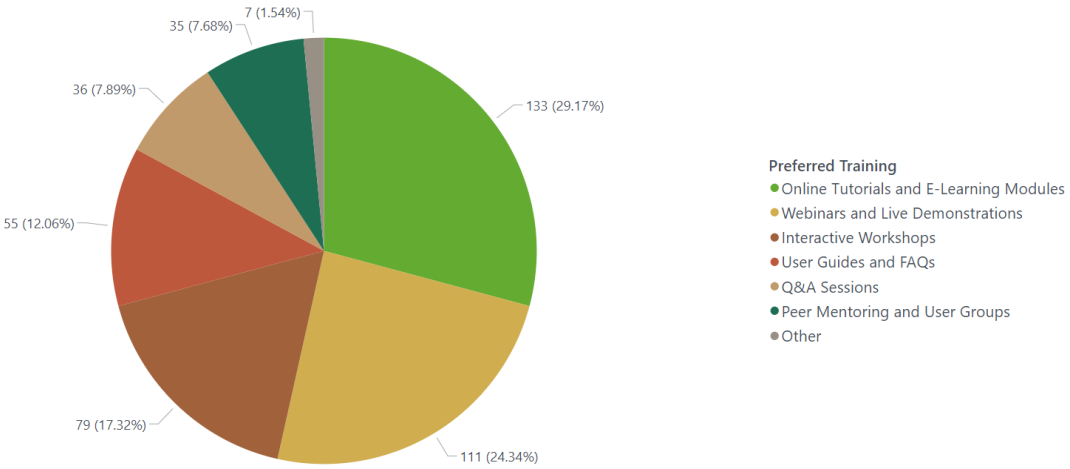


Figure 4.3.7: Survey results of type of training preferred

Analysis

In this section, the analysis of the data collected through the research methods is presented, focusing on the internal adoption and usage of Piab's AI chatbot, Pixie. The analysis aims to uncover patterns, trends, and insights from both qualitative and quantitative data. User interactions, feature utilisation, and performance metrics will be explored to identify key factors influencing the effectiveness and acceptance of Pixie within the organisation. Additionally, this section will highlight the challenges faced by users, their feedback, and recommendations for future improvements. By synthesising the empirical findings, the aim is to provide actionable insights and strategies to enhance the adoption and utilisation of Pixie across various departments in Piab.

5.1 Interview Analysis

A good spread of roles were showcased in the interviews, and the roles ranged from juniors to seniors to c-suite level. Many of the people interviewed were either in technical or sales roles at different levels within the organisation. The data exhibits a comprehensive mix of roles across technical, sales, operational, and managerial functions. This diversity supports a structure capable of handling various aspects of business operations from innovation and production to sales and people management. This provided a good spread of data, ensuring that the results would not be bias towards any particular group. Due to the needs and the goals of the business, it was not necessary to interview employees from warehousing and manufacturing. The sample range also captures newly employed people to employees that have been there for many years, perhaps showing recent growth or restructuring. The emphasis on sales and technical roles, especially in areas like Vacuum Automation suggests a strong focus on this division, and most interviewees were from this division. The information about roles proved to be useful and contributed to constructing the survey and deciding on dependent and independent variables.

Focusing on Pixie, a marketing campaign was in place for the launch of Pixie. Most of the candidates, about 50% found out about Pixie via email. The remainder of candidates found out about Pixie through meetings, word of mouth, or on their intranet. The most surprising one that few people were shocked to see is that Pixie suddenly appeared in their Team's meeting or web browser and they were thrown aback when they were told what it was there for. One

interviewee found out during a meeting that ” ... there’s some anonymous Joe in the teams call that will record the meeting.” Another interview mentioned that *”The noise about Pixie within the company has been substantial.”* Eventually, the usage slowly increased as people started to explore further and everyone had the basic understanding and knowledge about the capabilities. The study was focused on the Chatbot feature, and although everyone knew that it was integrated into MS Teams meetings to record the session and provide a summary, some of them have not explore it further. General capabilities enabling people to chat with Pixie and ask it general questions (similar to ChatGPT), creating summaries of meetings and documents, and using it for translations were known. Not everyone knew that they could upload their documents and ask Pixie to write summaries or search for information. All the people interviewed from France and China said they mainly use it for translations of documents and to translate emails to their preferred language.

”The noise about Pixie within the company has been substantial.”

”... there’s some anonymous Joe in the teams call that will record the meeting.”

5.1.1 Info from Usage Patterns

Exploring deeper, the way Pixie is being used varied depending on the roles and regions. The most common pattern that occurs from all users is the general prompt and search capability to look for products, documents, and general information since it also has access to ChatGPT.

The participants from France and China mainly use Pixie for translation of documents and translating emails for clients. Since Piab is a global company, other employees are also impelled to use Pixie to translate from over 10 languages.

In recruitment, CV documents are imported for populate evidence of key skills and development areas that are used to compare candidates and

5.1.2 Info from User Experience

A rating scale of 1-5 was used during the interviews, where 1 was a low rating and 5 was a high rating in terms of satisfaction with Pixie. The ratings that everyone provided were mostly positive and they always mentioned ways of improving it or ways that it could be better. Those

that gave it a low rating had very specific use cases that did not work for them in their role and were very technical people.

Some users mentioned that their satisfaction increased over time as they understood more about the tool and as it went through updates and enhancements. Several candidates mentioned that the interface is very similar to ChatGPT 4 so it was easy to use and understand the basics.

The biggest challenge that arose from all candidates is that the quality of information is not consistent (doubts with resources; inconsistent results; no limitations and too much info; does not have first hand knowledge). A couple of candidates mentioned that these types of technological tools develop and advance so fast that is difficult to catch up.

5.1.3 Info from Impact and Value

This explores insights from interviewees regarding the impact of Pixie on their work and productivity. Employees share instances where Pixie enhanced or hindered their tasks and suggest improvements to increase its value and effectiveness. These insights provide a clear view of Pixie's current capabilities and future potential within the organisation.

Many candidates responded positively when asked about the impact it has on their work, with time efficiency being the most prevalent and common value-add. It has saved time in many instances, where interviewees have responded with comments about summarised large PDF's, meeting summaries, and of course it has been valuable for translations. Using Pixie has allowed employees to be More organised when they meet with customers and more efficient in meetings. Being more efficient and having time saved in a few tasks opens up time to work on other activities.

5.1.4 Info from Feedback

This section of the interview was focused on capturing improvements that people would like to see, if they required further training and if Pixie was visible and accessible enough . Improvements

Almost everyone, except 5 people mentioned that further training would be beneficial to them, and would contribute to increasing their usage of Pixie. There were 2/3 people that requested physical printed training manuals so that they could read through it and learn how to gain skills to use Pixie at it's full potential. Others have learnt through trial and error and from prior knowledge of using Chatbots or from colleagues discussing how they have used it.

Some of the top-users have mentioned that it's all about the prompts and knowing what to ask Pixie, so they suggested training on how to *prompt* correctly in Pixie to get refined and efficient answers quicker, with less chatter. One user mentioned that " ... it's more about how to use it in a correct way and not something you can teach." Furthermore, people have asked for mandatory training sessions in the form of short video tutorials and promote it internally. A couple of people mentioned that it would be great if the training was split per department and wanted to know " ... what you can use Pixie for, not like what's the major AI trends."

"... it's more about how to use it in a correct way and not something you can teach."

5.2 Function specific insights into results

In order to gain a deeper understanding of the results from the qualitative and quantitative study and the impact on key departments (functions), we combine initial data collected and processed at the start of the study with survey responses. This approach helps to build function-specific insights into the major themes identified in the qualitative study.

It is essential to have a view of the number of respondents per function and the amount of messages sent per department, from the respondents whom have used Pixie. The data shown in Figure 5.2.1 is an overview of the respondents per function and the messages sent, which totals 251 and 42368 respectively.

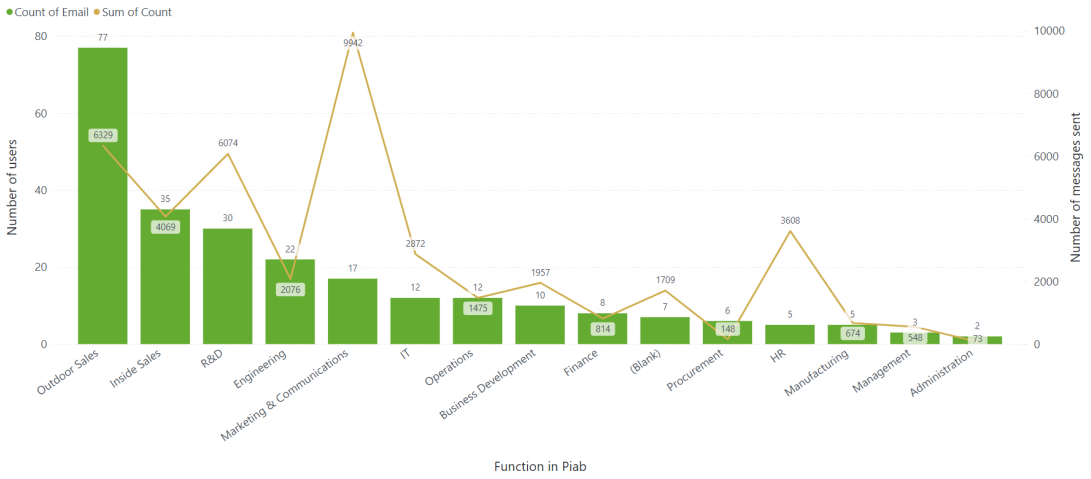


Figure 5.2.1: Respondents in survey per function and chat messages sent

When comparing data usage with survey results, focusing on the initial target function groups, clear patterns emerge. Figures 4.1.2 and 5.2.1 reveal a strong positive correlation between the

percentage of respondents using Pixie and the percentage of messages sent within different functions.

For instance, Marketing & Communications boasts high engagement, with 78% of messages sent by survey respondents from this function. Of the total Marketing & Communications user base, 47% participated in the survey. Similarly, Inside Sales shows 73% message usage, with 46% of the total Inside Sales population represented in the survey. IT demonstrates 66% message usage and a 52% survey response rate from its total user count. These functions exhibit a clear link between Pixie usage and communication activity, suggesting that the survey captured data primarily from 'champion' Pixie users. HR also stands out with notable engagement, sending 49% of messages and having the highest survey response rate at 56% of its total user base.

Interestingly, Manufacturing has a high percentage of messages sent (66%) despite a relatively low percentage of respondents (21%), suggesting high usage intensity among fewer users. Conversely, Management shows a lower percentage of messages sent (18%) despite a moderate percentage of respondents (33%), indicating potential under-utilisation of Pixie.

In conclusion, the correlation between the percentage of messages sent and the percentage of survey respondents using Pixie indicates a potential for user segmentation. A high proportion of messages sent relative to the number of respondents suggests that the survey may have captured data primarily from 'power users'. Conversely, a low percentage of messages sent in relation to respondents might indicate a more evenly distributed user base. Further investigation into user behaviour, potentially through detailed usage logs or additional surveys, could provide deeper insights into Pixie adoption patterns and the characteristics of power users.

5.2.1 Feature utilisation

The data offers an overview of Pixie utilisation across diverse organisational functions. It underscores the tool's adaptability to varying departmental needs, demonstrating how different teams harness Pixie uniquely to boost productivity. Table 5.2.1 presents a breakdown of Pixie usage by function group, expressed as a percentage of survey respondents who are active users. For example, the high engagement with features such as the General Search Function, Content Creation & Drafting across multiple departments emphasises Pixie's versatility in supporting both information retrieval and creative endeavours. The widespread use of Meeting Summaries and Document Summaries further highlights the tool's value for enhancing documentation and meeting efficiency. Overall, the data suggests that Pixie is seamlessly integrated into the daily

routines of various functions.

Function ID	Comparison of products	Document Summaries	General search function	Meeting summaries	Product information search	Content creation	Translations and language support	Other
Inside Sales	12%	7%	36%	26%	36%	36%	52%	10%
Outdoor Sales	21%	18%	37%	39%	40%	43%	38%	6%
R&D	14%	35%	62%	19%	38%	43%	35%	11%
Engineering	32%	16%	40%	24%	60%	40%	40%	12%
Marketing & Communications	29%	24%	53%	29%	35%	76%	59%	18%
Operations	0%	13%	53%	33%	20%	40%	33%	13%
Management	33%	100%	67%	0%	0%	33%	67%	33%
Business Development	33%	33%	58%	33%	33%	50%	42%	8%
IT	8%	42%	58%	83%	0%	67%	25%	25%

Table 5.2.1: Utilisation of Pixie per function group

Based on the provided data, several anomalies and inconsistencies in Pixie utilisation can be identified across different functions. Management exhibits a notably high reliance on Document Summaries at 100%, while IT demonstrates unusually frequent use of Meeting Summaries at 83%. Inside Sales heavily utilises Translations and Language Support at 63%, and Marketing & Communications frequently employ Sentence, Content Creation, and Drafting at 76%. Conversely, Operations exhibits no usage of Comparison of Products and relatively low utilisation of Product Information Search at 25%, while Management does not employ Meeting Summaries or Product Information Search at all. IT also demonstrates no use of Product Information Search, and Inside Sales exhibit low engagement with Comparison of Products at 14%.

Unexpected patterns include lower-than-expected usage of Document Summaries by Marketing & Communications at 24% and high usage of the General Search Function by R&D at 77% and Operations at 67%. Outdoor Sales show a higher reliance on Meeting Summaries at 48%, which might indicate a need to track client interactions. Engineering and Outdoor Sales exhibit relatively balanced usage across most features, suggesting a well-rounded application of Pixie. Investigating these deviations can help identify specific needs, barriers, or opportunities for better training and feature awareness across different functions.

5.2.2 Productivity

Productivity was measured purely on a subjective basis. In the survey, different ranges were given as options and when the data was analysed, the upper range values were used. In the case of the ”2+ hours”, it was rounded up to 2.5 hours of time saved per week. In total, about 221 hours of work have been saved per week by the 251 survey respondents. When aggregated across the entire company, the average time saved for all Pixie users rises to an impressive 692 hours per week. On average, an employee is expected to work 7 hours a day. Keeping that in mind, the hours saved per week is equivalent to around 20 FTE’s.

The project focus was to look at specific departments within Piab Group, so that insights can be drawn out to help increase usage and adoption, thus exploring options that would make these teams more efficient. Figure 5.2.2 shows the hours saved for the 9 chosen departments and the number of chat messages they have sent in the time window for this research.

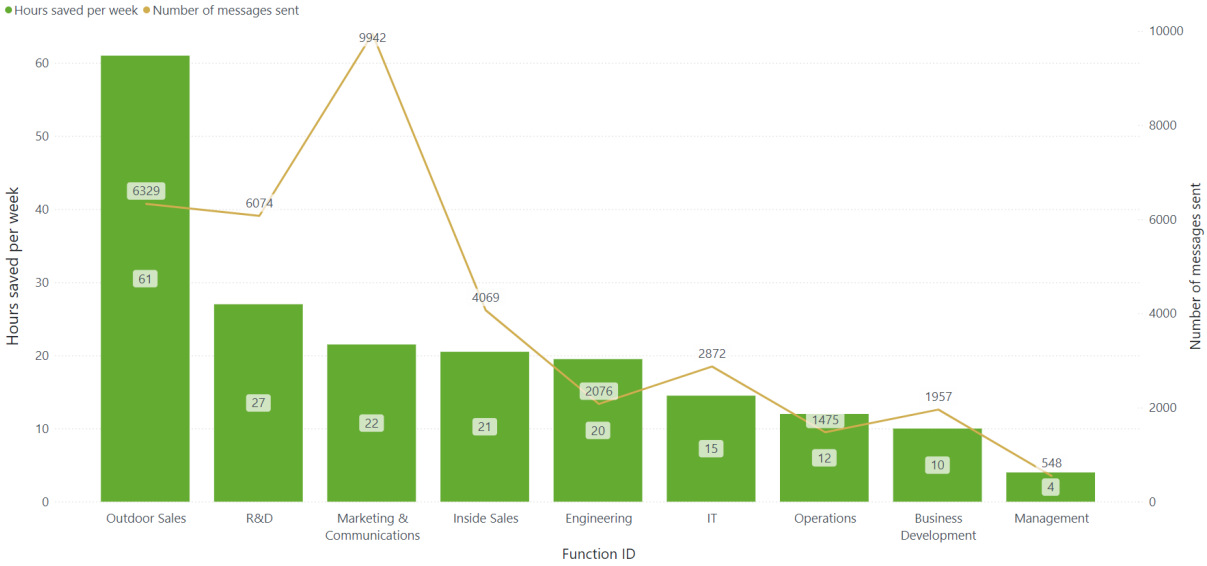


Figure 5.2.2: Productivity of key teams within Piab Group

5.2.3 Feature requests

It is to no surprise that the most requested feature is image reading/image creation. Not only has this request been high in the quantitative study, but many of the interviewees also requested this feature (see Figure 4.3.3 in previous chapter). From the survey, sales, marketing and engineering had the most requests for image reading/image creation. The second most requested feature is live translation within teams and the important part to note about this feature is that geographically, China and France have the highest votes, each with 10% and 17% respectively, followed by Sweden with 13%.

When looking through some of the comments where respondents provided their own opinion that was provided in the list, it was interesting to see that some of them requested features that were already incorporated within Pixie. Since many sales representatives formed the bulk of the respondents, there were many requests relating to sales roles such as information on competitors, Integration with SalesForce, and better search capabilities for products and/or components. Some of the requests were up and coming and a few of them mentioned specific requests for integration with role specific tools such as CAD, Power BI, and ERP.

5.2.4 Challenges

The list of challenges that were provided in the survey came from the deep analysis performed on the results from the qualitative study, research, and informative conversations with employees in the office during the study. Many suggestions were given for improvements together with some of the challenges that were encountered. There were a total of 10 different challenges and an option to specify ”other” challenges that were not mentioned. As shown in Figure 4.3.6 from the previous chapter, that just under 50% of the issues were due to 3 causes - Generic results, inaccurate answers and Pixie automatically entering meetings. The third challenge was soon resolved and an option to ask Pixie to join meetings was integrated into the chatbot. A lot of feedback from employees, and technological development needs to take place to continuously evolve and upgrade Pixie to resolve the two biggest challenges.

Given that Outdoor Sales and Inside Sales make up a substantial portion of the survey respondents, it is evident that their feedback heavily influences the overall perception of Pixie’s performance. For example, the issue of ”Pixie giving too generic results to questions” accounted for 20% of respondents coming from Sales (Outdoor and Inside), highlighting a critical area for improvement in providing more specific and relevant answers tailored to the needs of the sales teams

Amongst the other functions that were prioritised for this research, marketing, R&D and IT stand out as well. Marketing reported significant issues with accuracy, and this could indicate the need for a more tailored Pixie to support campaigns and communication strategies. R&D also highlighted major challenges related to the relevance and accuracy of Pixie. Pixie’s algorithms and search features could be enhanced and trained better to improve the precision of technical and research related queries to address these concerns.

These insights underscore the importance of addressing the specific challenges faced by the sales teams, as their feedback represents a significant portion of the user base. By focusing on improving the accuracy, relevance, and usability of Pixie for these key functions, the organisation can enhance overall user satisfaction and efficiency.

Furthermore, Figure 5.2.3 illustrates the reasons provided by several respondents from different functions who have not used Pixie at all. It is evident that there is a perception among some employees that the tool is irrelevant to their specific roles. However, this contradicts the experiences of other colleagues within the same functions who have found Pixie valuable. This inconsistency suggests that Pixie’s usage might be driven by individual needs and preferences

rather than strict job requirements.

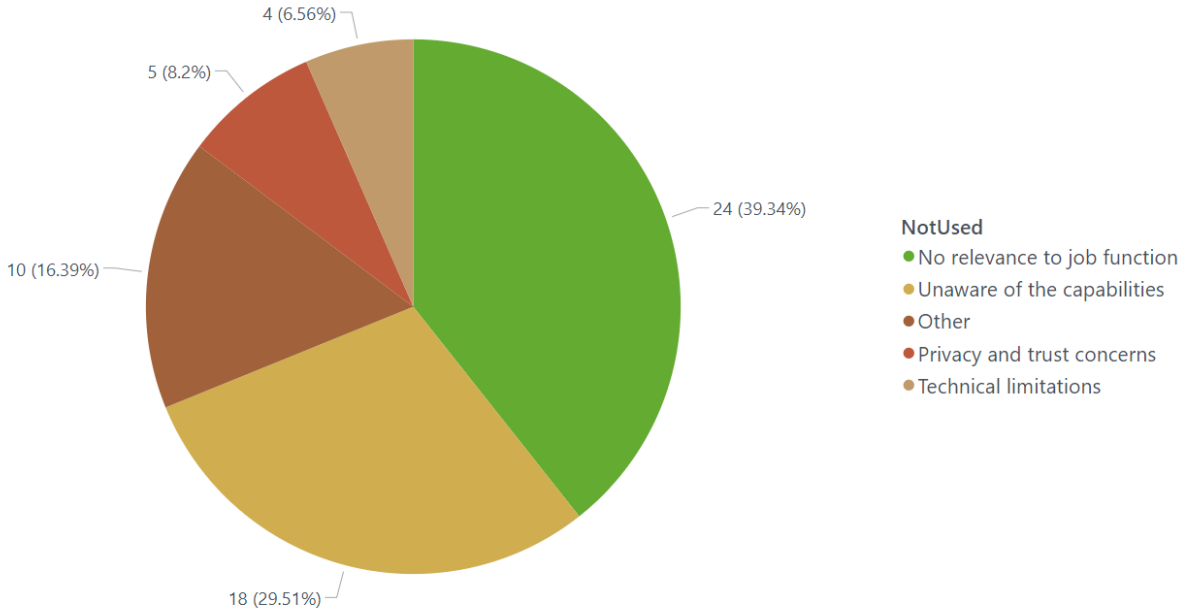


Figure 5.2.3: Challenges with Pixie

Discussion

This section begins with a concise overview of the study's methodology. Following this, the findings presented in Sections 4 and 5 are analysed and interpreted. The researchers' interpretations and responses to the research questions (RQs) are then discussed. Subsequently, the practical implications of the findings for Piab and other organisations are explored. Finally, the study's limitations and potential avenues for future research are outlined.

6.1 Overview

A qualitative and quantitative study was performed. The qualitative study recorded responses from 24 participants through interviews. The interview questions were governed by following a standard interview guide with open ended questions designed for this study. It had questions related to internal awareness of the AI tool (Pixie), usage experience & habits, purpose for usage, challenges faced during usage, new feature expectations and training. All the interviews were then coded to capture keywords and observe patterns in the experiences and perspectives communicated through the interviews.

From this, key findings from the qualitative study were narrowed down namely, like Popularly used features, Frequently occurring challenges and Feature Requests. These findings from the responses would only reflect the perspective of the employees interviewed within the company, which is a very small percentage of the total number of employees and users in the company. Since a larger sample was required to test and confirm on the findings from the qualitative study, a survey was designed with questions that tested the findings, forming the basis of the quantitative study.

The categories of questions were similar to the interview guide but had options that respondents could choose from. These options essentially represent the patterns observed during the interviews. The results from the survey, discussed in section 4.3 further confirmed most of the findings from the qualitative study in a quantified manner. Specific analysis of the qualitative and quantitative results are done in section 5 and derived answers to the RQ's will be discussed in the sections below.

6.2 Interpretation of findings

In order to discuss the findings, inferences and interpretation from it and its relation to the RQ's, the discussion will be organised as a structured answer under each RQ.

6.2.1 Answer to RQ 1

What drives the need for AI chatbots in organisational settings and what key features of AI chatbots enhance their effectiveness and value within an organisation?

The following are several potential drivers that discuss the need of AI chatbots in an organisational setting.

1. **Personal Assistance to employees**
   Upon closer analysis of the results from qualitative study, it was observed that the popularly used features, usage purpose and testimonials pointed towards the AI chatbot playing the role of a personal assistant to the employee, addressing most needs, day-to-day and role based challenges that the user may have and being available 24/7 to the employee as all they have to do to access it is logging into a website with their employee login. This then contributes as one of the major benefits of introducing an AI chatbot in the workplace. It serves as a contribution of the organisation towards the well-being and empowerment of employees. Mentioned below are some of the testimonials of users/employees that referred to Pixie as an assistant.

   As shown in Figure 4.3.1, the purposes of usage identified in the qualitative study were also included in the survey. While some purposes were more popular than others, overall, they all relate to the personal assistance that employees seek. Figure 4.3.4 shows that the features rated most useful in the survey—Summaries, Content drafting & structuring, and Translations—align with the purposes associated with personal assistance, confirming findings from the qualitative study.

2. **Automation of routine tasks**
   Automating routine tasks is another driver that the AI chatbot Pixie helps with (like drafting emails, searching for information, translating emails or creating presentations). This was observed and inferred through the quantitative study that, features that were ranked by many users as most useful are the features that automate routine office tasks and 47% of the users use the tool on a weekly basis and 29% use it on a daily basis. That is

more than 75% of users that make use it monthly, providing a means to be more efficient by automating tasks. When this number is scaled up, and should all users be considered, it is a significant portion of the company that is saving time through the use of Pixie.

3. **Keeping up with Technology**

   It is important for organisations to keep up to date with the latest trends in technology. This is in order to explore the relevance, possible benefits and understand its application withing their industry and ecosystem early on. At Piab, it is believed that AI is a ”pivotal force” that is reshaping industries and redefining how we work and interact with information. It is one of their strategic pillars for innovation and competitive edge and aligns with their core values ”Explore, Commit & Lead” (Piab Group, 2024).Jöhnk, Weißert, and Wyrtki (2021) conducted an interview study with AI experts on Organisational AI readiness factors. They highlight that exploring AI technology is important for organisations in order to capture the full potential of technology early on. They mention that it would help an organisation improve their decision making, strategic alignment, innovate and build necessary capabilities so they future proof their business and retain competitive edge.

4. **Fruitful Investment & Savings**

   AI Chatbots have the potential to perform and deliver tasks at a lower cost compared to the regular alternatives that organisations used before AI was introduced. One such example is discussed here. For organisations like Piab that have offices and customers in several countries, speaking different languages, their employees sometimes have to work with documents in languages they do not know. This raises a requirement for professional translations of the documents to the organisation’s business operation language (In Piab, it is English). Before Pixie was introduced, all document translations were done through an external agent that charged a minimum of 80-90 € for each document translation. An average of 1500 files are required to be translated in a year. With translation through Pixie, they are able to save a lot of this expense and gain more value for their investment in Pixie as they get a lot of other features too along with the ability to translate. Through the survey results, it is understood that translations is one of the most popularly used features and purpose for using Pixie.

5. **Data collection & analysis**

   From 3.2.1, it can be understood that the qualitative and quantitative study of the thesis relied on Sana AI’s ability to capture usage frequency of Piab employees using / not using

Pixie. This helped in the study to understand the level of use and adoption of AI chatbots at Piab and identify the increasing trend in both. The future potential of capturing usage data is also immense depending in the evolution of the tool at Piab and Sana AI.

The future potential of capturing usage data is immense, as it can evolve alongside the tool itself at Piab and Sana AI. Data collection not only provides insights into current usage patterns but also helps anticipate future needs and improvements, thereby fostering continuous development. There is definitely a lot more data that can be sifted through and more insights that can be obtained, depending on the needs and goals of Piab. One aspect that the team at Piab explored was the up-votes and down-votes that users sent through as feedback. This data and information is critical to improving and updating Pixie, and to ensure user satisfaction. It also shows that employees are being heard and creates a good company culture.

Moreover, Pixie's popular document summary feature was highlighted as one of the most beneficial tools in the workplace. The ability to read, analyse, and summarise structured data has proven to be invaluable, as indicated by survey responses and interview feedback. This reinforces the importance of robust data collection and analysis in understanding and enhancing the practical applications of AI tools like Pixie within an organisation.

6. **Scalability**

   Explained in section 4.1, combined analysis of the usage data from Sana AI and FTE data from Piab, was done in Power BI. It can be understood from the graph shown in figure 4.1.2 in that section that Pixie is used across all function groups in Piab. The usage frequency of course varies based on the size of the function group and relevance and applicability of the tool specifically to that function group. Apart from the research data, it is quite evident that Piab has managed to scale the software so that it integrates with their existing IT systems and workflows. This includes being compatible with several other software applications and platforms used within the organisation. It is important to keep in mind that effective integration will ensure that chatbots will enhance productivity without causing disruptions. Scalability is not just about handling more users but also about continuously innovating to meet evolving needs. Regular updates and enhancements based on user feedback are crucial for maintaining relevance and effectiveness. Engaging with users to understand their experiences and pain points can drive meaningful improvements. This study has provided valuable insights into user-experiences and additional feedback is always welcome. This goes to show that AI

chatbots are scalable in an organisation setting and can be used by most functional groups & roles.

7. **Data security and Privacy**
   An obvious argument against having an exclusive AI chatbot for employees of an organisation is to have the employees use commercially, freely available chatbots like chatGPT 4o or sponsor subscriptions for employees to advanced chatbots like chatGPT 5 and other attached products. The problem with such solutions is that the data derived from usage of commercially available chatbots by the employees might be subject to secrecy or competitiveness of the organisation and might be required to be kept private. Using commercially available chatbots would mean that the organisation would be exposing their data to competitor companies and organisations and would not have complete control over the ownership of the processed data. With AI chatbots like Pixie that are exclusively designed for use in and/or by organisations and their employees, the data and following derivatives comes under the control and viewing of only the associated organisation. This solution offers better control & privacy of the data to the organisation and its employees (Sana AI, 2024). An organisation can have ensure such a solution for an AI chatbot either through outsourced development of the chatbot or by developing an internal chatbot by themselves.

6.2.2 Answer to RQ 2

Which determinants affect the successful adoption and utilisation of AI chatbots in organisational settings and how can organisations overcome these challenges to maximise the benefits of AI chatbots?

From the results of the quantitative and qualitative study conducted, it was understood that there are three major determinants that affect and influence adoption of the Pixie. These determinants can also be considered as the influencing factors when considering application of other AI based office software tools

1. **Training and related enhancements** From the results in the qualitative and quantitative study, it can be understood that, training was a requirement for the employees as there was varying exposure and comfort with using AI chatbots. Discussed in section 4.3.7, it was understood that the participants of the survey required training to help with getting familiarised with the tool, thereby increasing adoption. Improving training and its effectiveness would therefore lead to increased adoption through the following pointers.

- **Visibility of training material**

  Discussed in section 4.3.7, it was observed that more that 50% of the employees weren't aware of the availability of training, but expressed interest in using the material if it's made available. Hence, improving the visibility of the training material to the employees through different channels (Company portal / Intranet, Common folders, Emails, Open discussions, etc.) would go a long way in improving the adoption of the tool and any clearing any perceived difficulty of adopting and using the tool in a potential user's mind.

- **Multilingual Support**

  Described in Section 4.2.3, and supported by the survey results, a need was observed for the training material on using Pixie to be available in the local languages of most Piab office locations. Since the training material was mainly video tutorials, a subtitle addition to the videos in the local language seems sensible. It should also be noted that, this recommendation is a result of the observation made from both studies. Piab follows the policy of English as its corporate language. The training videos are available in English. But in order to facilitate all employees to be able to access the material, going the extra mile with providing multilingual support would remove any barriers to using the training material.

- **Interface based updates, training and assistance**

  Since Sana AI conducts regular updates to its interface and new features are introduced, the user interface changes along with these updates. This means that the training material would need to be updated according to the latest version, or some form of pop-up or prompts within the interface would have to be integrated to help users keep up with the changes and new features. Difficulty with keeping with the changes and updates was a point highlighted and observed from the studies conducted. Among the highlights of some participants in both studies was the need for a simpler and user-friendly interface. It was understood that, with better training material and/or user assistance provided by Sana through its interface, the tool should get easier to use.

2. **Removing adoption barriers**

   With the adoption of any tool comes obstacles that tend to slow down adoption. Removal of these barriers can make the path to adoption of the tool much smoother. Hence, observed and perceived barriers to adoption of the AI chatbot at Piab was observed and

listed below:

- **Capability awareness**

  In chapter 5, from the analysis discussed in section 5.1, and backed by results from the survey discussed in 4.3.1, it can be understood that one of the barriers to adoption is low awareness and understanding of Pixie's capabilities and how to use it effectively, amongst non-users and infrequent users. The sure shot way to do away with this barrier is through above mentioned training and knowledge sharing amongst employees, between high users and infrequent and non-users.

- **Relevance to job role**

  Discussed in section 4.3, under subsection 4.3.1, it can be understood that the other barrier to adoption is the relevance of Pixie to the user's job role/function. While there are roles that don't have much requirement for Pixie and it's capabilities (Roles in Manufacturing and warehouse for example), they account for less than 20% of the total count of roles at Piab. But from the data processed in PowerBI, there are non-users in user potential roles too. Moreover, from the interviews in the qualitative study, some users also share the view of Pixie having low direct relevance to their particular job role. From observing the capabilities and potential features offered in Pixie, the software integration feature that lets users direct Pixie to perform actions in other connected applications would be potential direction to move in for increasing relevance to job roles. This would require closely understanding the applications that different job roles use and would benefit from automation and integration with Pixie, provided that it is feasible.

- **Cultural resistance to AI tools**

  The other barrier that was observed during the interviews was the cultural resistance to using AI tools. It is to be noted that this was observed to be a minimum but hold back some users from adopting the tool. The resistance was either in the form of not trusting the tool to do activities they are capable of doing by themselves or in the form of not using a tool they don't understand or know well.

Discussed below are determinants and potential drivers of utilisation of Pixie.

1. **Features with positive reception**

   Driving forward any updates and improvements that can further the performance of the following features is sure to drive usage upwards as they are the most popularly used and most useful features. This is understood from the results discussed in section 4.2,

subsection 4.3.5 and the analysis done in 5.2.

- Search functionality
- Written content creation
- Translation
- Summary feature

2. **Role specific tool integration**
   From the discussion under RQ1 in this section, it can be understood that by improving software integration and automation in Pixie, it would potentially improve relevance to the job role, which can improve both adoption and drive usage upwards.

3. **Image reading & Image generation**
   From the discussion of the results in the qualitative study in section 4.2, the results from the quantitative study in subsection 4.3.4 and the analysis in subsection 5.2.3, it can be understood, development directions towards bringing forward the image reading, processing and generation feature can majorly drive usage forward due to high demand and potential for contributing positively to different job roles.

4. **Improvement in critical performance factors**
   From the discussion and analysis of the results in subsection 5.2.4, it can be understood that two major challenges that need to be addressed are the return of generic results by Pixie and the inaccuracy of results. This corresponds to two critical performance factors of Pixie: Reliability and Accuracy. Developmental work towards improving Pixie on these two factors, improving the knowledge base of Pixie and its contextual understanding of the prompts given by the users would improve the trust and perceived quality of the answers given by Pixie thereby improving its usage.

Conclusion

The aim of this thesis was to investigate and explore the user experience of Piab's AI chatbot, Pixie, from a subjective standpoint through various research methods. The primary focus was on understanding the internal adoption and usage patterns, features, and overall performance of Pixie within the organisation. The research questions guiding this study were: (1) What drives the need for AI chatbots in organisational settings and what key features of AI chatbots enhance their effectiveness and value within an organisation? and (2) Which determinants affect the successful adoption and utilisation of AI chatbots in organisational settings and how can organisations overcome these challenges to maximise the benefits of AI chatbots?

The findings from the study revealed several key insights. Firstly, the need for AI chatbots in organisational settings is driven by their role as personal assistants that can automate routine tasks, enhance productivity, and keep the organisation technologically advanced. Features such as document summarising, content drafting, and translation were found to be highly valued by users. Additionally, the ability of AI chatbots to provide cost-effective solutions and facilitate data collection and analysis was highlighted as a significant advantage.

Secondly, the determinants affecting the successful adoption and utilisation of AI chatbots were identified. Training and related enhancements were crucial in improving familiarity and comfort with the tool. Removing adoption barriers such as low awareness of capabilities and perceived relevance to job roles was also essential. Overcoming cultural resistance to AI tools and improving critical performance factors such as reliability and accuracy were necessary to build trust and drive usage.

Overall, the study provided a comprehensive understanding of the factors influencing the adoption and usage of Pixie within Piab. The insights gained can guide future improvements and strategies to enhance the effectiveness of AI chatbots in organisational settings. By addressing the identified challenges and leveraging the key drivers of adoption, Piab can maximise the benefits of Pixie and ensure its successful integration into the daily workflows of its employees.

7.1 Implication of findings

The findings, directives and pointers provided by this study can be used by Industries in a similar position and market as Piab pr by firms looking to implement and/or their internal AI

chatbots that are used by their employees only. Another potential group that this study could provide benefits and perspective to is AI software developers. Firms and individuals involved in the research and development of RAG model AI chatbots could use the findings from this paper for understanding potential use cases and outcomes of AI chatbot implementations and its effect and perception in an office environment and as an office tool.

7.2 Limitations of the study

This study suggests potential directions of development through the indication of drivers of usage and adoption. The exact technical solution to enable work in those directions are not researched into. The study focuses on the perspectives of a RAG model based AI chatbot in an office environment, by internal employees. It does not focus on the common users perspective of using AI chatbots with similar capabilities to Pixie. Although there is a high potential of applying Pixie's capabilities in a customer facing role, that aspect is not researched in this study. This study only focuses on the perspective of employees of one firm, Piab, particularly using one AI chatbot tool called Pixie. So although, it considers the perspectives of a wide pool of users about how it is to use an AI chatbot, the perspectives are only from employees of 1 company an AI chatbot. A better study would consider perspectives of employees of different firms, using different versions of the chatbot or entirely different chatbots.

7.3 Future research directions

A potential future direction of this research would be to understand how the AI chatbot can be made more relevant, specific and beneficial to the different roles inside the company, Piab in this case. Keeping focus on the particular use case of Piab, another direction of research could be to study the potential of Pixie being implemented in a customer facing role and the corresponding feasibility. Understanding the different parameters and features that the chatbot should have in order to be an efficient customer communication and relations tool and the cost benefits of this direction could be the potential aspects to study further.

Another direction could be to perform the same study done in this research, but consider the use case of another firm with a different set of employees/users using a different RAG AI chatbot. The study can be further strengthened by considering multiple use cases in different firms that are part of varying industries in the market, for example: Studying the use cases of a firm in a mechanical and manufacturing industry, fashion industry, software industry etc).

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Appendix - Contents

A Interview Guide ii

B Project Plan iv

C Additional Power BI Charts v

D Interview Coding vii

Interview Guide

Introduction (brief intro on us, interview, and study)

- Do you consent to us using your replies in our study?
- Do you wish to be anonymous in our study?
- Would you be okay with the interview being recorded?
- Can you describe your role at Piab?

Background information

- What do you know about Pixie and its capabilities?
- How did you come to know of Pixie ?
- How long have you been using Pixie ?
  - What was your first impression?

Usage patterns

- How often do you use Pixie? (Daily, Weekly, Monthly?)
- What do you use Pixie for?
- Can you describe the last few times you used Pixie?
- What prompts you to use Pixie ?
  - Did you know that Pixie has GPT4 capabilities, and you can use it for more than just your work?
- How related is Pixie to your current job role?

User experience

- On a scale of 1, 3 & 5 how would you rate your satisfaction with Pixie and its results?
  - (Follow up question) Why?
- What feature of Pixie do you find most useful? How workflow has been impacted?
- Have you encountered any challenges using Pixie? Please describe them.
  - Can you describe times it has failed you?

- Is there anything you wish Pixie could do, which it cannot do currently?

Impact and Value

- How does Pixie impact your work and tasks at the job? Describe please.
- How does it impact your productivity, and can you share an instance (Succes or not)?
- What can be improved in Pixie for it to add more value to your work and role?

Feedback

- What about Pixie can be improved to improve user experience?
- Do you feel additional trainings / tutorials on using Pixie would be required for you and your colleagues to utilize Pixie to its fullest potential? Elaborate on what training.
- Do you feel Pixie should be more visible and accessible to all the employees compared to how it is now?

Closing

- Thanks
- Explain how these inputs will be used further in the study
- Do you have any questions for us?

Project Plan

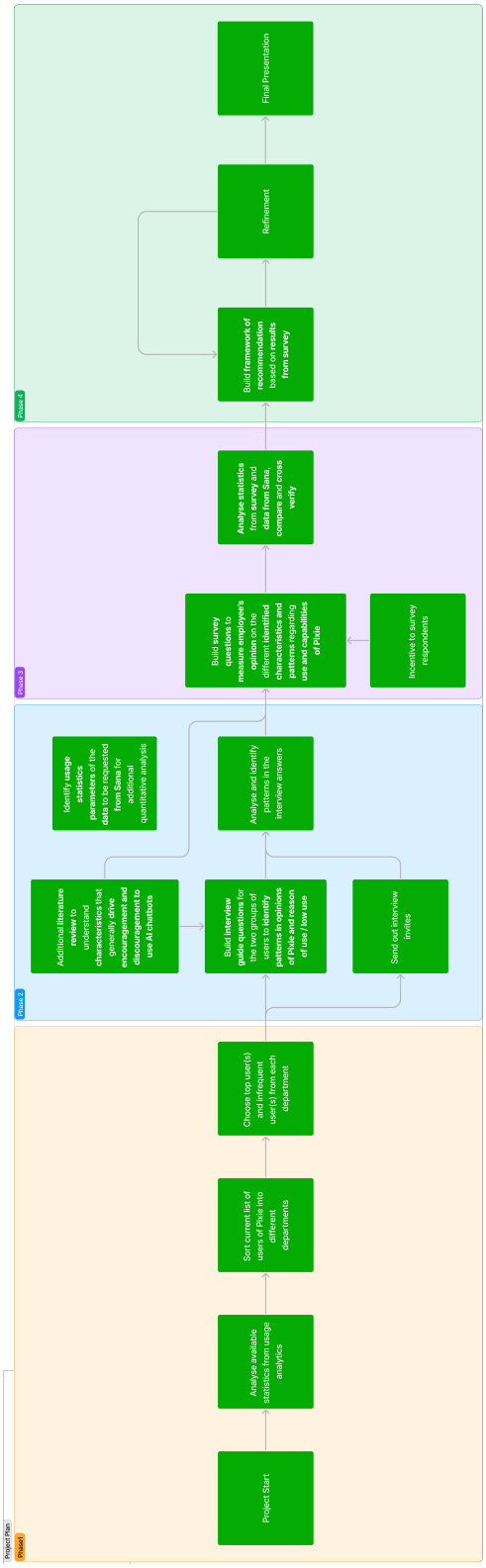


Figure B.0.1: Project plan flow

Additional Power BI Charts

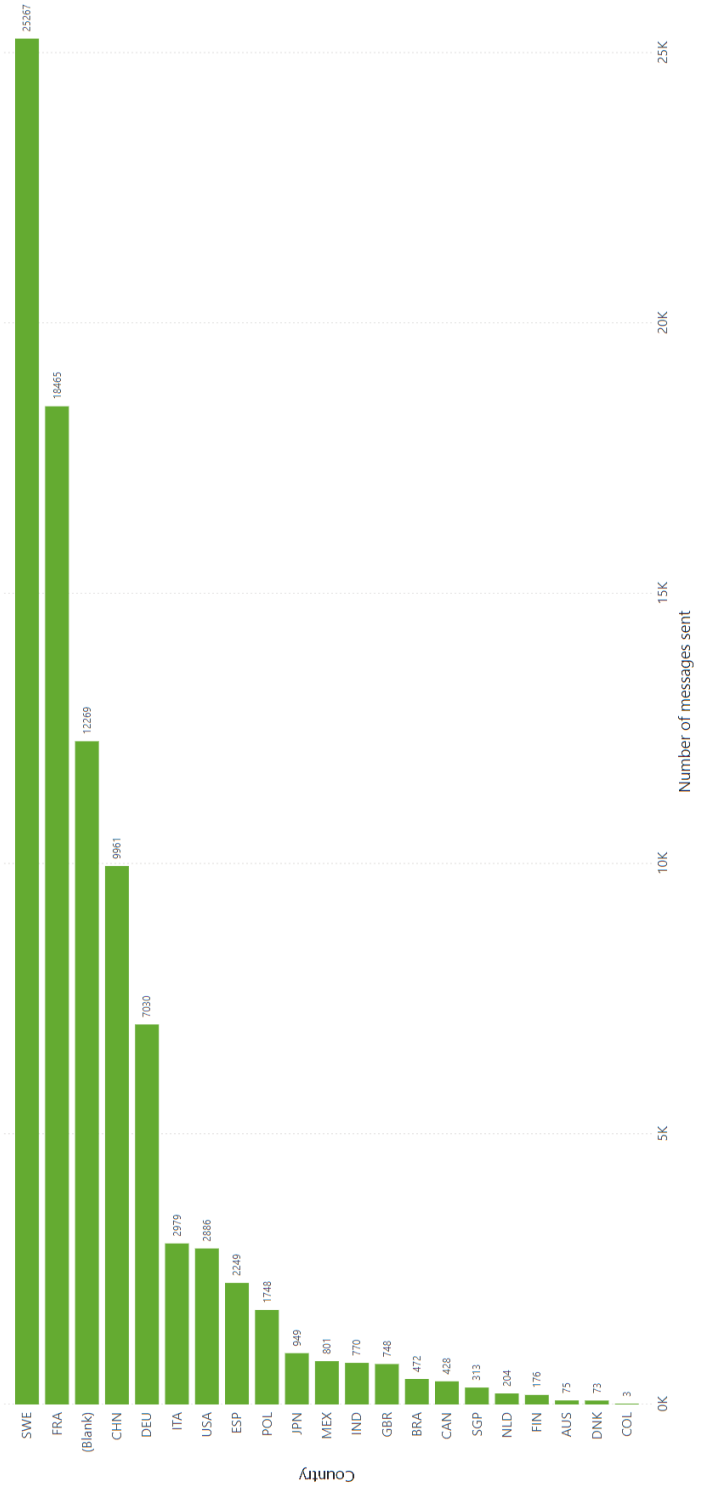


Figure C.0.1: Number of messages sent per country

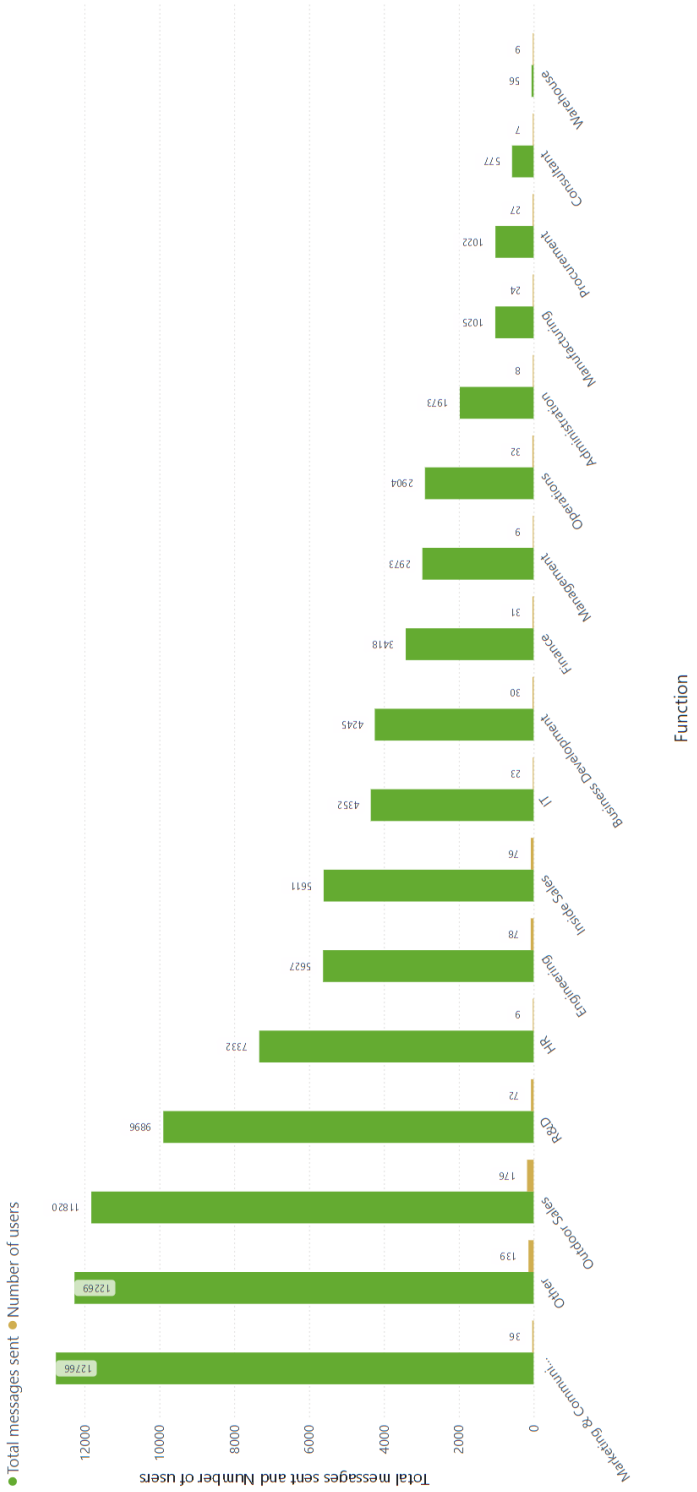


Figure C.0.2: Messages sent and number of users per function

Interview Coding

Basic themes	Count
Accessibility	14
Additional comments	14
Challenges	101
Feature requests	91
Knowledge of capabilities	2
Learning material	20
Perspective	77
Policy & Regulations	9
Source of introduction to tool	26
Training & Development	37
Capabilities	7
Knowledge gap	11
Adapting to new technology	3
Accuracy	2
Utilisation	154
Performance	57
Productivity	52
Efficiency	22
Role	42
Reliability	2
User Interface	3
Understanding	1
Integration	1
Frequency of use	23
Tool Competition	1

Table D.0.1: Basic themes and observed occurrences

The count refers to number of statements made by the interviewees regarding the corresponding basic theme.

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